Chedam: work log

Cumulative record of every working day, newest first. This copy covers up to Sunday, October 11, 2026. Owner: Sreya. Baseline: Master Specification v1.0.

Days in this log

- **2026-10-11**: P5 Connect and enrich built and deployed; testing round 3 fixed; web store moves onto the Pi
- **2026-10-10**: P2 finished (steps 6-11) and deployed; P3 started
- **2026-10-09**: Testing feedback round 1 (T2): fixes and changes built, tested and deployed
- **2026-10-08**: P1 step 9 deployed; P1 step 10 (gate) built, deployed and its automated parts run
- **2026-10-07**: Fixes from Sreya's testing; P1 step 3 deployed; P1 step 4 offline selling built, tested and deployed; till opens offline; P1 step 5 receipts and printers built and deployed; P1 step 6 returns, exchanges and store credit built and deployed; P1 step 7 shelf labels built; manual test sheet; Pi health, backup and power-cut checks; P1 step 8 import and export built and deployed; P1 step 9 money and audit reports built
- **2026-10-06**: P0 gate: power pull ×10 passed after two real fixes (Wi-Fi lost at boot, confirmed writes lost); untrusted-certificate warning
- **2026-10-05**: P0 steps 1-9 built and deployed; gate: 3-device login, update + rollback, backup + restore, memory under load passed: hub base, schema + event log, access, devices, forgot PIN, client app; work log started
- **2026-10-03**: New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed
- **2026-10-02**: P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

2026-10-11 (Sunday, October 11, 2026)

P5 Connect and enrich built and deployed; testing round 3 fixed; web store moves onto the Pi

Summary

Sreya: "continue to P5, deploy to the Pi; next, slight changes in plans: e-commerce hosting using the same Pi for the business." P5 is planned (`docs/P5-mini-spec.md`, `docs/sprints/Sprint-5.md`, 9 steps) and built the same way as P2-P4: each step gets its own hub test suite (run once), a client build, a deploy to the Pi and a commit. The full suites, Pi test runs and browser walkthroughs wait for the next test week. The web store change is recorded as **DL-214**: the shop will run on the store's own Pi through an outbound tunnel, replacing option A of DL-114 (a static host). It is the next sprint after P5.

Decision: the web store on the Pi (DL-214)

- The shop runs as a **separate process under its own user** with a read-only copy of the catalogue that the hub refreshes. It never touches the hub's database.
- It is reached through an **outbound tunnel** on the owner's domain (Cloudflare Tunnel). No port forwarding is needed; the TELUS modem has none anyway.
- Online checkout uses the processor's hosted page (Square, the same account as the terminal). Paid orders come back as pick-up orders with stock reserved.
- Rate limits apply, and the shop stops itself before it can slow the tills. Production hardware is the Pi 4.
- Accepted trade-off: the shop is down whenever the store's power or internet is.
- Manager remote access (FR-12.06) uses the same tunnel and moves to that sprint.

P5 steps built and deployed today

Step	What was built	Tests	Deployed
1. Card processor, terminals (DL-215..217)	Home → Card processor (owner): paste the Square access token (sandbox or live). The hub checks it with Square and keeps it sealed. Pick the location; pair a Square Terminal with a device code and link it to a till (one terminal per till). The common payment interface (`lib/payproc.js`) with the Square adapter (`lib/payproc_square.js`); a fake Square for tests and the dev hub (`tools/devhub/fake-square.mjs`)	27	`dev-cae6dd1`
2. Linked card payments (DL-218..220)	Card on a linked till → Send to the terminal; the till waits for approved / cancelled / timed out; cancel from the till; Key it in instead (standalone, P5-d). The hub fills the card type, last 4 and Square's payment id itself. A checkout is used once, for its sale and amount only. A card paid while the sale was refused, or the page reloaded, comes back for the same sale. Training never charges a card	26	`dev-8b7ed3f`
3. Linked refunds, fees, payouts (DL-221..223)	Returns and voids of a linked sale: Back to the card through Square (Interac on the terminal: the customer taps). The return or void names the refund and the hub checks it; nothing is refunded twice. Paid checkouts that no sale took are listed on Sales for managers to refund. Hourly: Square's fee per payment (card fees and the P&L use it) and its payouts (Fees and statements)	25	`dev-1e4b309`
4. Cloud drive, off-site backup (DL-224, 225)	Home → Cloud drive (owner): Google Drive or OneDrive, signed in with a device code (google.com/device, microsoft.com/devicelogin); the app registration (D2) is entered on the screen. The backup helper leaves each verified encrypted backup in an outbox; every 5 minutes the hub uploads it (curl, token in a 0600 header file) and keeps the newest 14 sets in the cloud. A fake Google/Microsoft (`tools/devhub/fake-cloud.mjs`) for tests	21	`dev-c722dea`

5. Owner remote view (DL-226..228)	Every 15 minutes an encrypted summary (today vs last week, the month, tills open, who is clocked in, what needs attention, stock, last backup, hub health) goes to the cloud drive as `chedam-summary.enc` (AES-256-GCM with the remote view key). The owner enrols their own phone at the store; away from it, `#away` signs in to the cloud drive, downloads the summary and decrypts it with WebCrypto, with "last updated". "Forget enrolled devices" makes a new key. The phone's reader is the same code the hub test runs	16	`dev-2333c56`
6. Import everything, templates (DL-229..231)	Import other records: customers (only with the store's confirmation that they agreed to a loyalty program; points carried over), vendors and clients (with a contact), opening stock (a receiving "Opening stock" for products without stock), open bills and invoices (one "Opening balance" line, no tax), staff (owner only; person + employee record, SIN encrypted, never in the job record). All or nothing, in the import log. Built-in column layouts of Square, Shopify, Lightspeed, Clover, Loyverse, QuickBooks, Xero, Wave and Zoho exports are recognised; the store saves its own templates	36	`dev-fd4ee2c`
7. Reference data (DL-232..234)	Datasets as read-only SQLite files with the hub's code (units, Canadian sales tax by province, a shelf-life starter guide, the 97 HS chapters), outside the store's database; read with `sqlite3` (on the PC a Node stand-in). Online, one item at a time: Open Food Facts by barcode, Bank of Canada rates. Suggestions only, with Use buttons: product name/French name/size from the barcode, shelf life, HS chapter, exchange rate; province rates on Tax; Help → About lists every source and licence	19	`dev-f1d16d9`
8. Remote support (DL-235, 236)	Home → Remote support (owner): the support person's public SSH key; "Let support in" for 1-8 hours. Root helper `chedam-support`: user `chedam-support`, keys with OpenSSH `expiry-time`, sudo for the session only, an outbound Cloudflare quick tunnel, a timer that ends it even if the hub is down; End now; sign-ins shown; every session recorded. `cloudflared` is not installed yet (the screen says so)	15	`dev-41a0eaf`

After the selling, return and fee changes, the sales (75), offline (35), returns (43), recon (13), statements (13) and backups (30) suites were re-run, and they still pass.

Found and fixed

- **Backups did not keep the hub's own keys.** A backup saved the database and the HTTPS certificate authority, but not `pb_data/chedam-hr.key`. After a restore onto a new card, the employees' SIN and bank details (P4) could not be read. `chedam-backup` now saves `pb_data/chedam-*.key` beside each backup (`.keys`, encrypted with the backup key). The restore puts them back and keeps the current ones aside (DL-216).

Failed cases and fixes (all in the new suites, before deploying)

- Step 5: the summary could not be uploaded on a hub that had never made a backup: `pb_data/backup` did not exist yet for the scratch file. `cloud.js` now creates the folder first.
- Step 6: the test created a product through an endpoint that does not exist; it now creates it with the product import. The P1 import/export suite found that the new P5 tables were in no group of the "Export data" chooser; they now are (and hidden tokens stay out).
- Step 6: the Import screen needed `catalogue.edit`, which would have kept the accountant from importing open bills. It now opens for anyone with one of the import permissions; the hub checks each import.
- Step 7: route helpers declared at the top of `reference.pb.js` were not visible inside handlers (the known PocketBase rule); moved into `lib/reference.js`. Products have no SKU field, so the SKU match was removed.
- Step 8: Python's `pwd` module does not exist on Windows, which stopped the helper's self-test on the PC; it is imported only where it exists. The "Add a key" form would have replaced every key (the screen only sees masked keys); the hub now takes add / remove as well as replace. A test picked the first start request instead of the latest.
- The fake Square ignored `fake.down = true`: the server read `st.down`. Fixed with a getter and setter in the fake.
- A checkout timeout setting of 0 seconds was read as "not set" (300 s), because of `||`. Fixed: only a missing value means 300.
- A linked-payment test sold more towels than the sample stock has. The test now receives stock first.
- The refund test read the refundable amounts from `returns/find`, a short list. They come from `returns/sale/`

`{id}`, the test was fixed.

- Cloud uploads made a new "Chedam" folder on each upload and kept using an expired access token. The code used a stale copy of the link record. Fixed: it re-reads the record; on a 401 it refreshes the sign-in and tries once more.
- Svelte warned that `returnId` (Returns screen) was not reactive, so the refund chooser would keep the old return id after "another return". It is now `$state`.

Commands

```
PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/step52-payproc.test.mjs        # and step53..55
cd client && npm run build
bash hub/scripts/deploy.sh --sample-data
python tools/testing/build_test_sheet.py --p5
node tools/devhub/fake-square.mjs     # dev hub with CHEDAM_SQUARE_BASE=http://127.0.0.1:8097
node tools/devhub/fake-cloud.mjs      # dev hub with CHEDAM_CLOUD_BASE=http://127.0.0.1:8098
```

Needed from outside

- Square developer account, sandbox access token and a Square Terminal (D1): steps 1-3 for real.
- Google and Microsoft app registrations (D2): steps 4-5 for real, and the P2 email fallback.
- Sreya's OK to install `cloudflared` on the Pi (remote support now, the web store next).
- Sreya's OK to download the large reference datasets (Open Food Facts Canada subset, CBSA tariff, nutrient files, FoodKeeper, ITC-HS).

The P5 manual test sheet is `docs/testing/Chedam-Manual-Tests-P5.xlsx` (56 checks).

P5 gate (automated part)

`docs/gate/P5-gate.md`. 60 hub suites on the PC pass (about 1,700 checks), after one count fix in `step2` (3 new permissions). On the Pi: 46 suites pass on throwaway hubs (1,323 checks). Event log on the Pi's hub: 699/699, integrity ok. Pi load (5 tills, stress pace, 4 minutes): no lost or duplicated sales, memory peak 161 MB (as P4), payment to receipt p95 1.36 s (DL-107). The load test found one refusal ("open a larger pack?" for a case with only loose cans); fixed under DL-240.

Commands:

```
PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/run-all.mjs
bash tools/pitest/run-on-pi.sh step3-access ... step51-accountant-exports step57-import-more
ssh chedam 'sudo -u chedam-hub python3 - /opt/chedam/pb_data/data.db' < tools/gate/event-coverage.py
ssh chedam 'bash -s start' < tools/loadtest/pi-loadtest.sh   # then sales-load.mjs through the 18099
```

Testing round 3 (Sreya, Android): fixed and deployed

Details and answers: `docs/testing/T3-feedback-2026-10-11.md`; decisions DL-237..245; retest sheet `docs/testing/Chedam-Retest-2026-10-11.xlsx` (18 checks).

- **Messages would not open** (any conversation with 2+ messages): the hub reversed PocketBase's own record list in place, which repeats entries, and the screen stopped on the repeated message ("each_key_duplicate", seen in the browser at phone size on the dev hub). Fixed; the messages suite now has a back-and-forth conversation. Chat style: people row, last message time, reactions on tap.
- **Pings:** online list refreshes every 15 s; Take back shows "Called off by …" on every device it went to.
- **Cashier view:** only Sell, Products, Time clock, Roster, Customers, Messages, Pings, Me / own time sheet; other addresses go Home.
- **Sell beyond stock:** on by default, no message on the till; urgent task + inbox alert to owner, managers, stock staff.
- **Products:** brand, supplier, stock added with a received date, Duplicate, Below zero (printable).
- **Vendors:** how we order (portal address, account, user name; never a password); tabs for products, consignment,

bills and invoices, orders; **Buy: best supplier** with draft order and quick-order file. Filling a portal's cart by itself is not done (explained in the feedback doc; possible per vendor through EDI or an API).

- **Scale:** optional till scale (Web Serial), untested without hardware.
- Also: stale "Coming in a later phase" tiles removed; variant option order fixed (it changed between runs).

Failed cases while fixing: the offline and two other suites sometimes failed to start their throwaway hub on the PC (a port still closing); they pass when run again alone. The sales and variety suites expected short stock to be refused; they now test both settings.

2026-10-10 (Saturday, October 10, 2026)

P2 finished (steps 6-11) and deployed; P3 started

Summary

Sreya's instruction: develop non-stop, deploy each step to the Pi, and test everything at once at the end of P4 (one week reserved). So today each step got its own hub test suite (run once, to catch my own mistakes), a client build, a deploy and a commit; the full suites, the Pi test runs and browser walkthroughs wait for the test week. **P2 development is complete** (steps 1-11), deployed as **`dev-33c1ca9`**. The P2 manual test sheet has **98 checks** (`docs/testing/Chedam-Manual-Tests-P2.xlsx`). P3 (Buy and spend) is planned and started.

Step	What was built	Tests	Deployed
6. Dashboard and sales insights (DL-139, 140)	Home → Dashboard: attention list first, today vs last week, month vs last month, 30 days with last year, margin by category, P&L month to date from Chedam's records, stock health; Reports → Insights: weekday × hour heatmap, best and slowest sellers, sell-through, year over year. Charts: Apache ECharts, only the parts used	19	`dev-ad21e2e`
7. Tasks, checklists, reminders (DL-141..143)	Manual tasks (who, due, urgent, link to a record); opening/closing checklists with who ticked what and a late task; handover notes with read receipts; licences, permits and insurance with expiry reminders; wrong-storage tasks from product temperature ranges	30	`dev-9342032`
8. Messages and announcements (DL-144..146)	Everyone channel, groups, one-to-one; @mentions, product links, photo/PDF, reactions, read receipts; live through PocketBase realtime, delivered only to members; announcements that need confirming, and who has / has not	31	`dev-786fa57`
9. Pings and the till overlay (DL-147..149)	Pings to one device, all tills, phones, back office, everyone; templates and replies; over any screen without losing a sale; urgent full screen with a beep until confirmed; unanswered urgent pings go to managers after 5 minutes and open a task	19	`dev-7e4d4b4`
10. Inbox, end-of-day report, email fallback (DL-150..152)	Each person's inbox: the end-of-day report (owner and chosen people), urgent alerts, per-kind choices; deadlines; unread items past the deadline are marked once for email (sending waits for the Google/Microsoft client IDs)	19	`dev-b050138`
11. Help (DL-153)	The **?** opens the help page of the current screen; 19 pages with steps, tips, search, working offline; training videos stored on the hub	7	`dev-33c1ca9`

Notes

- Unfinished background jobs from yesterday: the three listed were log watchers that had already delivered their results (all Pi suites passed); nothing needed redoing.
- New top-bar buttons: **?** help, inbox, ping, messages (with counts).
- Failed cases and fixes during the day (all found by the new suites before deploying): (1) sell-through counted base units twice (a sale line's `base_qty` is already the line total); fixed. (2) A dashboard test expected staff not to see stock value, but staff see costs by design (DL-72); the test was wrong. (3) Route helpers declared at the top of a hooks file are not visible inside handlers (PocketBase runs each handler on its own); the new route files require their helpers inside each handler. (4) A ping escalation test raced the minute job; it now checks the outcome.
- Permissions added in P2 today: `checklists.manage`, `documents.manage`, `announcements.manage` (managers). 45 permissions in all.

P3 Buy and spend: steps 1-10 built and deployed

Step	What was built	Tests	Deployed

1. Parties (DL-154..156)	Vendors and clients (one record can be both): contacts, terms, currency, tax and importer details, documents; a communication log whose follow-up dates become tasks; exchange rates by day	21	`dev-3b5083d`
2. Vendor prices (DL-157..159)	What each vendor sells us (their code, unit, cost, currency, minimum, lead time, preferred); price list import from CSV/Excel matched by their code then barcode; comparing vendors in CAD per base unit; better prices	18	`dev-25f25ae`
3. Purchase orders, reorder (DL-160..162)	Draft → send (PDF) → receive partly or fully through the normal receive action (incoming stock, differences flagged with a task) → close / cancel; reorder point + order-up-to max make draft orders per vendor	24	`dev-6932331`
4. Bills, invoices, payments, vendor returns (DL-163..166)	Bills, vendor credits, invoices, client credits with currency, taxes, due dates, photos; payments incl. applying credits; aging; statements (PDF); a bill from a received order (three-way); returns to vendors take stock out and wait for their credit; vendor performance	26	`dev-71a3a4f`
5. Receive from a bill photo, landed cost (DL-167..169)	The phone reads the bill (Tesseract served by the hub, pdf.js for PDFs), the hub matches lines (learned, their code, barcode, name), review grid, totals and tax checks, the open order checked; freight, duty and brokerage spread by value into item costs; the matches are learned; the bill can be recorded	17	`dev-8d534c0`
6. Expenses and petty cash (DL-170..172)	Claims with receipt photos; approve / reject; pay back from the till (a pay-out), petty cash, cheque, e-transfer, next pay; duplicate and missing-receipt flags; the petty cash box with top-ups and counts	19	`dev-11a52ab`
7. Layaway, special orders, quotes, house accounts (DL-173..177)	Customer orders with reserved stock and deposits at the till (counted in the drawer and the card total); ring up at the till with the deposit as a payment; quotes become orders, invoices or sales; house accounts: "On account" makes an invoice within the credit limit	21	`dev-b86bef9`
8. Product variety (DL-178..180)	Bundles take their components out of stock (sale, offline, return); variants made from a parent's axes, the till asks which; serial / IMEI per unit, never sold twice, warranty lookup	18	`dev-d2dfe8a`
9. Delivery orders, consignment (DL-181, 182)	Delivery-app orders typed in from the platform's tablet: platform prices, stock reserved, picked up = a sale paid by the platform; consignment products owed to the vendor per unit sold (returns taken back), outside the stock value, billed per vendor	15	`dev-570390c`
10. Recall trace, feedback (DL-183..185)	Lots traced to sales and members; recalling writes off what is left and blocks the lots (FEFO skips them); feedback from a kiosk tablet or the receipt link, report with the average	17	`dev-23f86f2`

New permissions: `parties.view`, `parties.manage`, `purchasing.view`, `purchasing.manage`, `finance.manage`, `expenses.approve` (51 in all). The P3 manual sheet (`docs/testing/Chedam-Manual-Tests-P3.xlsx`) has 66 checks. The sales, offline, returns, pricing, stock, markdown, printing and dashboard suites were re-run after the selling and stock changes in steps 7-10 and still pass.

Failed cases and fixes in P3 so far (all found by the step suites): sample vendor codes collided (ids share a prefix; a counter now); new vendor products defaulted to inactive (a yes/no field reads as "no", not "not given"); an empty price list cost read as $0 (now "no cost"); stored variant keys come back sorted, so existing variants were not recognised (compared sorted now); a bill test hit stock limits before the credit limit (test data); the accountant could not turn a quote into an invoice (permission widened for that action); an offline bundle sale wrote its movement on the bundle instead of its components (one movement per component now); a newline inside a string literal broke a build; a recall test assumed the wrong lot would be sold first (FEFO picks the earliest expiry: the test now gives the recalled lot the earliest).

P2-P4 gate (automated part)

All 57 hub suites pass on the PC and 48 on the Pi itself after one test fix (step3 read only the first 500 events; the P4 migrations now write more at start-up). Event log on the store hub: 586/586. Pi load with the new features: 5 tills at stress pace, 0 errors, payment to receipt p95 1.17 s, memory peak 161 MB. Books screens after ~300 sales: 1.2-2.6 s each on the Pi Zero; a busy month will be slow (noted to watch). Backup test-restore waits for the backup drive. Report: `docs/gate/P2-P4-gate.md`; Sreya's week: `docs/testing/P2-P4-test-week.md`.

P4 People and money: steps 1-10 built and deployed

Step	What was built	Tests	Deployed
1. Employees and leave (DL-186..188)	Home → People: records per person (job, pay, address, emergency contact, documents); SIN (checked) and bank details encrypted, owner-only, masked, revealed on request (logged); managers see no pay; everyone sees their own record (Me); leave ledger with monthly accruals, the BC sick grant, time taken, adjustments with a reason	22	`dev-1a18249`
2. Time clock (DL-189..191)	Home → Time clock: clock in/out and breaks on your device, or on any device with your name and PIN; the hub's clock; BC alerts before a missed meal break or overtime (screen, inbox to the person and managers); missed punch-out alerts; Shifts for managers with the regular / overtime / double split, flags, fixes with a reason keeping the original, missed shifts added	35	`dev-3dbcb29`
3. Roster (DL-192..194)	Home → Roster: weekly grid, drafts, publish (each person told in the inbox), copy last week, overlaps refused, BC warnings; swaps offered, taken, approved; labour cost against sales per day (totals only)	30	`dev-f46b115`
4. Time sheets (DL-195..197)	Home → Time sheets: per pay period (setting), hours split by BC rules, leave taken, flags (late, no meal break, not rostered, no-show, overtime, no punch-out, fixed); approval with a note, locks the punches; reopen with a reason	20	`dev-1e57c28`
5. Payroll (DL-198..200)	Home → Payroll (the accountant): runs per ended pay period from approved time sheets and salaries; overtime, sick pay, vacation pay, unpaid leave, expenses paid back with the pay; CPP, EI and tax **entered** from PDOC (never calculated); finalise → time sheets paid, pay stubs (People → Me, with year to date); payment recorded; T4 boxes and ROE data	24	`dev-8002a53`
6. Bank (DL-201..203)	Home → Bank: accounts; CSV (RBC-style, TD without a header), OFX, QFX imports without duplicates; automatic matches (vendor and client payments, expenses paid back, payroll, cash deposits); card batches matched by hand with the fee; categories; cash deposits from tills; reconciliation to the statement balance	24	`dev-a0655d3`
7. Fees and statements (DL-204..206)	Card type at the till; card fees by type (estimated, then actual from the bank); delivery-platform payout statements checked against Chedam's orders (and matched to the bank); vendor statements checked against the bills	13	`dev-561fc92`
8. Books (DL-207..209)	Home → Books: the journal derived from every money record by fixed rules (~60 accounts, renameable, mappable to the accountant's codes); trial balance; months closed in order with checks, then locked (BR-34); owner reopens with a reason	23	`dev-601cd07`
9. Statements and returns (DL-210..212)	Profit and loss (with the period before), balance sheet (balanced), cash flow; GST/HST return lines 101-113A; BC PST return with the commission; filed returns kept	13	`dev-1dbbc86`
10. Accountant exports (DL-213)	QuickBooks Online and Xero CSV: general journal, sales journal (one a day), payroll journal and register, expense claims, bank reconciliation	12	`dev-43f07a2`

New permissions: `hr.view`, `hr.manage` (owner), `timeclock.manage`, `roster.manage`, `payroll.manage`, `bank.manage`, `books.manage` (the last three for the accountant; 58 in all). New inbox kind: "My schedule". The P4 manual sheet (`docs/testing/Chedam-Manual-Tests-P4.xlsx`) has 82 checks. After the selling change in step 7 (card type), the sales, offline, delivery and bank suites were re-run and pass.

Failed cases and fixes in P4 (all in the step suites, before deploying): a time sheet test assumed today's pay period had not ended (it ends today, and its last day can be approved); a payroll test used an expense category not in the list; a test sold more towels than the sample stock has; a test expected a round 30% commission (30% of the gross rounds to 29.99%); the books test expected cash without the 2-cent cash rounding (the books were right).

P4 People and money: planned

`docs/P4-mini-spec.md` and `docs/sprints/Sprint-4.md`: 1 employees and leave, 2 time clock with BC break and overtime alerts, 3 roster, 4 time sheets, 5 payroll preparation, pay stubs, ROE and T4 data, 6 bank import, reconciliation, deposits, 7 card fees, platform payouts, vendor statements, 8 books and period close, 9 financial statements and GST/HST, PST returns, 10 accountant exports, 11 gate and the P2-P4 test week. Design choices to note: SIN and bank details encrypted, owner-only; deductions entered from the CRA calculator, not calculated (R4); the books derived from

Chedam's records by fixed rules; BC first.

P3 Buy and spend: planned

`docs/P3-mini-spec.md` and `docs/sprints/Sprint-3.md`: 1 parties (vendors and clients), 2 vendor products and price lists, 3 purchase orders and min/max reorder, 4 bills, invoices, payments, vendor returns, 5 receiving from a bill photo and landed cost, 6 expenses and petty cash, 7 layaway, special orders, quotes, house accounts, 8 variants, bundles, serial numbers, 9 delivery orders and consignment, 10 recall trace and feedback, 11 gate (with the test week).

2026-10-09 (Friday, October 9, 2026)

Testing feedback round 1 (T2): fixes and changes built, tested and deployed

Summary

Sreya tested on the laptop and the iPhone and sent a list of fixes and requests. Everything functional is built, covered by hub tests (**808 pass, 21 suites**) and checked in the browser against the dev hub. **Deployed to the Pi** with Sreya's go-ahead (see 6). The decisions are DL-109..113 in `docs/P1-mini-spec.md`. The retest sheet is `docs/testing/Chedam-Retest-2026-10-09.xlsx` (21 checks).

Request	What was done
"Saved" message on every screen that changes something	"Saved ✓" pops up at the bottom after every change the hub accepts (`client/src/lib/toast.svelte.js`, `components/Toast.svelte`, called from `api()`). Pricing, selling, printing and sign-in calls are left out
Remove one item from the cart	A × on every line (removing a line already existed, but only inside the line's edit panel). Totals update by themselves
Receipt for 15 s after a sale, then back to selling	Countdown "Back to selling in 15 s · Stay here"; any tap on the receipt area keeps it
Discounts per product, and on the whole sale, shown clearly	Per-line discounts already existed, also hidden in the edit panel. Now "Discount this item" under each line and "Sale discount" for the whole sale. Each discount is described ("10% off", "$2.00 off") and kept on the sale, so the sale screen and both receipts say which discount was for which product (DL-109)
Stock recent changes: sold vs received in different colours	Received/returned green, sold blue, damage/loss red, with a key; a voided sale shows "Sale voided"
Receipt reprint needs a manager	Reprints need `sales.approve` or a manager's PIN, on the network printer and the device's own printer; counted; who allowed it is recorded (DL-112)
Till number goes up at every open/close	A device keeps its number (Till 1, Till 2); the count of openings is the Z number (DL-110). On the Pi's data, all 4 tills become Till 1 (Z 1-4)
Labels on the iPhone: "'text/html' is not a valid JavaScript MIME type"	Cause found and fixed (see 2)
Labels on the laptop: one vape pod label with a different price	Cause found and fixed (see 3)
Product import from Excel or Google Sheets, columns shown first	The import already read Excel/CSV; now it shows the columns (required / needed to sell / optional, with examples) and gives an Excel or CSV template that maps itself
Reports: batch number generated, in order, no duplicates	`B-000001`, `B-000002`… given by the hub at reconcile time; unique index; a refused reconcile uses no number (DL-111)
Export: choose what	Products, stock, sales, tills/returns/refunds, people/devices, store settings, audit log; zip (CSV + JSON) or Excel workbook
Home button (C logo), business logo, fixed top bar	Noted for later (U1 in the sprint), as Sreya asked: after the functional work
Square/Clover terminal; web store on the Pi with the domain	Written up: `docs/proposals/card-terminal-and-webstore.md`; decisions needed (N2)

1. Steps and commands

1. Read the screens and hub code for each item. Found that line removal and line discounts already existed, but

only in the line's edit panel, so they were easy to miss.

2. Pulled the label batches from the Pi (read only): `ssh chedam "sudo -u chedam-hub sqlite3 -json /opt/chedam/pb_data/data.db \"select ... from label_batches\""` Batch 8 (2026-10-09 13:55) had 10 × Vape pods at $19.99 and the label data was right. Redrew it in Node with the app's own `labelsPdf()` and rendered it with PyMuPDF to find what looked wrong (see 3).
3. Built hub changes, migration `1791300015_p1_feedback.js`, client changes; new test suite `hub/tests/step20-feedback.test.mjs`; updated `step15-printing` (reprints) and `step18-import-export` (template, export groups).
4. `PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/run-all.mjs` → 21 suites pass, 808 checks.
5. Migration dry run on a **copy** of the Pi's database (`.backup` on the Pi, copied down, deleted from the Pi's /tmp): `pocketbase migrate up --dir=<copy> --migrationsDir=hub/pb_migrations --hooksDir=hub/pb_hooks` → applied; tills 1-4 get `register_no` 1; 4 audit entries by `system:migration`; batch index created. The copy was deleted afterwards.
6. Browser check on the dev hub (`chedam-dev-keep`, new launch config that keeps data): cashier sale with a 10% item discount and a 5% sale discount, card payment, receipt, 15 s return, reprint asking for Mira Manager's PIN (counted, approver recorded), "Saved ✓" on Categories, stock colours, import template downloads (CSV + Excel), Sales-only Excel export, till close and reconcile → batch B-000001.
7. `python tools/testing/build_test_sheet.py --retest` → `docs/testing/Chedam-Retest-2026-10-09.xlsx`.

2. iPhone label PDF error

- **Cause:** the PDF maker is loaded only when needed (a separate file of the app's build). When the hub got a new build (deploy) while the app was open on the phone, the new service worker deleted the old build's files from its cache. The page still ran the old build, asked for the old PDF file, and the hub answered with its start page (HTML). Safari then said "'text/html' is not a valid JavaScript MIME type".
- **Fix:** (a) `client/src/main.js`: on Vite's `vite:preloadError` the app reloads itself once (at most once a minute), which opens the new build; (b) `client/src/sw-template.js`: the service worker keeps the previous build's cache until the next update, so an open page can still load its parts.

3. Wrong price on a vape pod label

- **Not a data error:** all 10 vape pod labels in batch 8 carry $19.99.
- **Cause:** each label put the name at its top edge and the price at its bottom edge, with empty space between. On the sheet, a price ended up right above the next row's name. The vape pod label below "Paper towels 6 rolls" therefore read as $10.99, and with any print offset or scaling, that price lands on the next label. That makes exactly one vape label "wrong", as reported.
- **Fix:** `client/src/lib/labels/pdf.js`: name, details and price are one block centred in the label. Checked by redrawing batch 8 (before/after images in the session).

4. Found while testing

- **A 10% discount asked for a manager.** 10% of $3.99 rounds to $0.40, which is 10.03% > 10%. The limit now compares the percentage asked for (`pricing_core.discountPct`, hub and offline till). Test added.
- The first try at a test of migrating back on a copy got stuck at PocketBase's y/N question, and the second reverted PocketBase's own aux migration on a partial copy. Neither touched real data. Reverting is already covered by step 2's test, which reverts every Chedam migration including the new one.

5. Files

Area	Files
Hub	`pb_hooks/lib/sales.js`, `offline.js`, `pricing_core.js`, `printing.js`, `printing.pb.js`, `tills.js`, `reports.js`, `receipt_layout.js`; `pb_migrations/1791300015_p1_feedback.js`

App	`screens/Sell.svelte`, `Sales.svelte` (via PrintButtons), `StockItem.svelte`, `ImportExport.svelte`, `Reports.svelte`; `components/PrintButtons.svelte`, `Receipt.svelte`, `Toast.svelte`; `lib/api.js`, `toast.svelte.js`, `export.js`, `offline_price.js`, `import/template.js`, `labels/pdf.js`; `main.js`, `sw-template.js`, `app.css`, `App.svelte`
Tests	`hub/tests/step20-feedback.test.mjs` (new), `step15-printing.test.mjs`, `step18-import-export.test.mjs`
Docs	`docs/P1-mini-spec.md` (DL-109..113), `docs/sprints/Sprint-1.md` (T2, U1, N2), `docs/proposals/card-terminal-and-webstore.md`, `docs/testing/Chedam-Retest-2026-10-09.xlsx`, `tools/testing/build_test_sheet.py` (`--retest`)

6. Deploy (Sreya's go-ahead)

- `bash hub/scripts/deploy.sh --sample-data` → backup before deploy (integrity ok), version **`dev-c5587c9`**, hub healthy, migration `1791300015_p1_feedback.js` applied, no errors.
- On the Pi: `sqlite3 ... "select register_no, number, status from tills"` → all 4 tills are Till 1 (Z 1-4); `pragma integrity_check` ok.

7. Decisions (DL-114)

- Card terminal: link the processor the store already has (Square, Clover…); Square when the store has none. Stays in P5. The common payment interface (FR-3.22) comes with the first link.
- Web store: added to the plan (sprint table row 6), **option A** (confirmed by Sreya): Chedam publishes the shop to a static host, the hub stays private. The store owner manages the domain and DNS.

Next

1. Sreya: retest with the retest sheet on the laptop and iPhone (Android later).
2. Later: UI items U1 (Home button, logo, fixed top bar).

8. Retest passed; UI requests U1 built

- Sreya: all 21 retest checks passed.
- U1 (DL-115): the Chedam "C" in the top bar goes Home; the owner is asked in Store setup → Logo and receipt whether the logo also shows in the app (`business.logo_in_app`, migration `1791300016_p1_logo_in_app.js`), then it shows in the top bar and on Home; the top bar and each screen's header row (23 screens, class `screen-head`) stay fixed while scrolling; on phones Sell's buttons stay on one row.
- Checked in the browser (dev hub): logo upload with the question, logo in the top bar and on Home; Products scrolled 406 px with its header held under the top bar; phone width 375 px: no sideways page scroll, Sell header one row. Hub tests: 21 suites pass.
- Not deployed yet (needs Sreya's go-ahead).

9. P2 planning

- Read every P2 requirement from the Master Specification (FR-2.01-2.12, 3.14, 4.12, 5.06-5.12, 5.20, 6.16, 7.01-7.08, 10.10, 12.07, 12.11; BR-20, 21, 24, 40-42).
- Wrote `docs/P2-mini-spec.md`: 12 build steps (promotions first, then markdowns and staff discounts, customers and loyalty, reports, customer display, dashboard and insights, tasks and checklists, messages, pings, inbox and end-of-day report, help, gate), first design choices P2-a..g, open questions.
- Raised with Sreya: the spec says P1 is released and piloted before P2 starts, and P1's gate items (L01-L06, Q1, N1, H1) are still open. Build P2 alongside, or finish P1 first?

10. Deploy U1; decisions; receipts as PDF

- Deploy with Sreya's go-ahead: the first try failed at the copy step (`ssh: connect to host 192.168.50.101`

`port 22: Software caused connection abort`, nothing had changed on the Pi). Ping was fine a moment later, so it was a short Wi-Fi drop. The second try worked: **`dev-c80615e`**, backup integrity ok, migration `1791300016_p1_logo_in_app.js` applied, no errors.

- Decisions (DL-116): P1 pilot week skipped (the Pi has run since the start; production on a Pi 4); Q1 tax review, N1 real numbers and H1 printer tests at the end; P2 starts; the owner decides the loyalty settings (Q6).
- Receipts as PDF: "Save as PDF" on the receipt and return slip. It uses the hub's receipt layout in the browser (`virtual:receipt-layout`, `client/src/lib/receipt_pdf.js`), on 80 mm paper with a CODE128 receipt number; a second copy is a reprint (manager rule). The hub now includes `till_number` in sale and return views. Checked by rendering a sample PDF to an image, and in the browser (manager: saved as copy 2, counted). Tests: step13/15/16 pass, step20 18 pass.
- A test sign-in through the PIN pad in the browser went wrong twice (the pad submits by itself after 4 digits, so an extra OK sent an empty PIN). Dev hub only; switched to `tools/devhub/dev-login.mjs`.

11. P2 step 1: promotions and scheduled prices (built, not deployed)

- New: `hub/pb_hooks/lib/promotions_core.js` (pure engine, also on the till as `virtual:promotions-core`), `lib/promotions.js`, `promotions.pb.js` (endpoints + minute label job), migration `1791400001_p2_promotions.js` (tables `promotions`, `scheduled_prices`; sale line and sale fields; permission `promotions.manage` for managers), screen `client/src/screens/Promotions.svelte`.
- Changed: `lib/sales.js` (scheduled price, best deal, then the cashier's discount; deals stored on lines and the sale; uses counted), `lib/offline.js` (deals and scheduled prices in the offline pack; offline sales keep their deals), `client/src/lib/offline_price.js` (same engine offline), `lib/labels.js` + `labels/pdf.js` (price in force and "SALE" labels), `receipt_layout.js` + `Receipt.svelte` (the deal under its product), Sell (deal per line, deals in the totals, Coupon button), Home tile, access, retention, export groups.
- Decisions DL-117..123 in `docs/P2-mini-spec.md`. Sprint file `docs/sprints/Sprint-2.md`. Manual sheet `docs/testing/Chedam-Manual-Tests-P2.xlsx` (P01-P02 receipt PDF, P10-P23 promotions).
- Tests: new `hub/tests/step21-promotions.test.mjs` (54: every deal type, best deal, stacking, limits, days/hours, coupons, exact cents, hub quote/sale/receipt, preview margins, labels, scheduled prices, offline parity with deals, offline sale keeps its deals, module off). `step14-offline` loads the new engine; `step2` counts 39 permissions. **All 22 suites pass.**
- Browser (dev hub): created "Snack week" (10% off Snacks) with the preview (chips $3.99 → $3.59, cost $1.90, margin 47.1%), switched it on, rang up chips + chocolate + cola: deal under each snack and in the totals. A sale label with a deal rendered and checked as an image.
- Found while testing: `#promotions` was not a known address (opened Home); added to the session's address list. Svelte warned that a helper named `state` hid `$state`; renamed.
- One test expectation was wrong: uses count each application (4 items), not each sale (DL-120).

12. Deploy step 1; P2 step 2: near-expiry markdowns and staff discounts (built, not deployed)

- Deploy with Sreya's go-ahead: **`dev-47be3cf`** (promotions + receipt PDF), backup integrity ok, migration `1791400001_p2_promotions.js` applied, no errors. One hour later the hub's log had no errors from the new minute job.
- Step 2 (DL-124..127): migration `1791400002_p2_markdowns_staff.js` (settings `promotions.markdowns`, `sales.staff_discount`, both off; `stock_lots.markdown_pct`, `label_batch_items.markdown_pct`, sale staff fields). Engine: `markdownPct`, `markdowns` (lots in selling order), markdown as a deal. Hub: `markdownSegments`, minute sticker job, `/api/chedam/promotions/markdowns`, staff discount in `sales.js` with a staff-only PIN approval and the monthly limit. Offline: markdowns in the pack and priced on the till; staff discounts refused. Screens: Promotions (markdown and staff settings, stock marked down now), Sell ("Staff sale" with `components/StaffPin.svelte`), receipts.
- Tests: new `step22-markdowns-staff.test.mjs` (25), including stickers made by the minute job. **All 23 suites pass.**
- Failed cases and fixes: 1. The sample-data migration (P1) queues labels before the new `markdown_pct` field exists → "unknown field markdown_pct". The label queue now checks the field is there. 2. Engine: choosing the

deal with the biggest total saving gave 30% off both items ($2.40) where 50% markdown on one plus 30% on the other ($3.20) was better for the customer. Per-item deals are now chosen item by item; bundles only when they gain more (DL-125). Renaming slip: `single` was already declared later in the function (Node: "Identifier 'single' has already been declared"). 3. Test expectations: a deal's saving is rounded once for the line (2 × 25% of $5.49 = $2.745 → $2.75, not 2 × $1.37); a staff sale of 2 chips is $1.60 off, not $0.80. 4. Browser: saving the markdown settings reloaded the screen and threw away an unsaved change in the staff discount form. Each Save now keeps the other form's edits.

- Browser check (dev hub): markdowns on → "Sourdough loaf, expires 2026-10-12, 6 at 25% off"; Sell: chips (Snack week –$0.40), sourdough (Near expiry 25% off –$1.37), staff sale for Sam Staff with his PIN: $0.00 on items with a deal, cola –$0.15, "$99.85 left this month".
- Manual sheet: P30-P40 added (`Chedam-Manual-Tests-P2.xlsx`, 27 checks).

13. Step 2 deployed and tested on the Pi

- Deploy with Sreya's go-ahead: **`dev-d20c953`**, backup integrity ok, migration `1791400002_p2_markdowns_staff.js` applied, no errors.
- New: hub test suites can run **on the Pi itself**. `tools/pitest/run-on-pi.sh` starts a fresh throwaway hub on the Pi for each suite (`tools/loadtest/pi-loadtest.sh`: the Pi's PocketBase, the deployed hooks and migrations, the sample store, its own data folder on 127.0.0.1:8099; never the store's hub), opens an SSH tunnel (127.0.0.1:18099), and runs the suite from the PC in remote mode (`CHEDAM_TEST_REMOTE`, `hub/tests/lib/hub.mjs`). Suites that start the hub their own way stay on the PC (step2, 6, 7, 9, 10, 12, 15).
- Result: **16 suites pass on the Pi** (step3, 3b, 4, 5, 8, 11, 13, 13a, 14, 16-22); test hub memory peak 76 MB, Pi free memory lowest 181 MB.
- Failed cases and fixes: (1) `pgrep` is not in Git Bash, so the tunnel was not closed; the script now uses an SSH control socket and closes it on exit (the leftover tunnel process was stopped). (2) step5-status failed on the Pi only: the test pointed the hub's internet check at the PC's tunnel address, which the Pi cannot reach. The harness now gives the hub's own address (`t.self`); passes on the Pi and the PC.

14. Step 3 deployed and tested on the Pi

- Deploy: **`dev-00775f6`** (customers and loyalty), migration `1791400003_p2_customers_loyalty.js` applied, no errors on the store's hub. The new settings are in place and **off** until the owner sets them: loyalty programme, near-expiry markdowns, staff discount.
- Result: **all 17 Pi suites pass** (step3, 3b, 4, 5, 8 in the first run; step11, 13, 13a, 14, 16-23 in a clean rerun, 505 checks). Test hub memory peak 77 MB, Pi free memory lowest 175 MB. All 24 suites pass on the PC.
- Failed cases and fixes in the test tooling: (1) a stalled SSH tunnel (Wi-Fi drop) hung one suite for 46 minutes: requests in remote mode now time out after 90 s, the tunnel sends keep-alives, and each suite has a 15-minute limit. (2) "failed to upsert superuser: unique" on start: test hub starts left behind by stopped runs were still on the Pi; they were stopped, and the script now stops and retries once when a start fails.
- Next: Sreya's manual testing with `docs/testing/Chedam-Manual-Tests-P2.xlsx` (40 checks); then P2 step 4, promotion and loyalty reports.

15. P2 steps 4 and 5: promotion and loyalty reports, customer display (deployed)

- **Step 4, reports (DL-135, 136).** Reports has two new tabs. *Promotions*: each deal's times used, sales, units, sales before tax, savings given and (with cost permission) margin, against the same number of days just before; the "lift" (all units and sales of the deal's products, now and before); coupons by code; deals sold below cost flagged. *Loyalty*: members (all, joined, bought), members' share of sales and average sale, points earned / used / taken back / adjusted, the points customers hold and what they are worth in $ at the owner's value (a liability for the accountant), top customers (first name and last 4 digits only, managers only). Both export. 23 tests.
- **Step 5, customer display (DL-137, 138).** Pair a tablet, phone or stick PC as a "Customer display"; in Devices choose which till it shows. It shows the items with deals and "you save", the total, the member's first name and points, "To pay" while paying, then "Thank you!" with the change, and the store's name/logo when idle. No sign-in,

no buttons. The hub keeps only the latest screen per till in memory (nothing in the database or event log). A laptop till with a second monitor: Sell → "Customer screen" opens the display window, fed straight from the till, so it also works with the Pi off. 19 tests.

- Checked in the browser: both report tabs with sample data; a training sale on the till shown live on a separate display device (items, Snack week deal, $0.68 saved, total), payment ("Thank you!", change $3.40), back to "Welcome" after 15 s; the second-monitor window following the sale; the device manager's "Shows the sale of".
- Failed cases and fixes: (1) a test asked to delete a customer without the confirmation the hub requires; the test was corrected. (2) A test read a table name that does not exist (`event_log`; it is `events`), so it passed without checking anything; corrected and re-run.
- Deploy with Sreya's go-ahead: **`dev-a119768`**, migration `1791400004_p2_customer_display.js` applied, no errors; the live hub answers the new endpoints.
- Tests: **PC: 26 suites, 969 checks, all pass. Pi: 19 suites, 750 checks, all pass** (test hub memory peak 76 MB, Pi free memory lowest 178 MB).
- Manual checks added: P70-P76 (reports), P80-P86 (display); the P2 sheet now has 54 checks for Sreya's morning testing.
- Next: P2 step 6, dashboard and sales insights.

2026-10-08 (Thursday, October 8, 2026)

P1 step 9 deployed; P1 step 10 (gate) built, deployed and its automated parts run

Summary

- Step 9 (money and audit reports) deployed: **`dev-9e9ae0d`** (2026-10-07, 22:20), retention 6 years, integrity ok.
- **Step 10, the P1 gate:** tools to test sales under load, power cuts mid-sale and audit-log coverage; run on the Pi. The load test found two real problems, both fixed and deployed (**`dev-5859ec5`**, then **`dev-b0e54bb`**): 1. **Hub memory:** a list of 500 records with costs took the hub from about 50 MB to 230-416 MB (hard limit 220 MB): the Stock and Products screens ask for products 1,000 at a time. 2. **Return costs visible to cashiers** through the generic API.
- Results: 5 tills, no lost or duplicated sales; memory levels off at ~99 MB at store pace (peak 138 MB); power cut mid-sale 3/3 (311 sales, nothing lost or half-saved); restore test 29 tables; every record in the audit log.
- **Not met:** payment-to-receipt under 1 s at the 95th percentile on the Pi Zero (median 0.45 s; 1.1-1.7 s when tills pay at once). Recorded as DL-107 for Sreya to decide.
- Manual test sheet now 109 checks (`docs/testing/Chedam-Manual-Tests-P1-1-10.xlsx`), with the gate checks that need Sreya (L01-L06).

1. Gate tools

File	What
`tools/loadtest/sales-load.mjs`	N tills: scans, quote, sale (cash/card), returns, voids, discounts, reports, audit log; an offline burst from all tills at once, each sent twice. Checks: every confirmed sale exactly once, unique receipt numbers, stock = sum of movements, Z = sales, an audit entry per sale, payment p95
`tools/loadtest/pi-loadtest.sh`	Throwaway hub on the Pi (+ `POOL=` for the JS engine pool) with a 2-second memory sampler
`tools/powertest/sale-writer.mjs`	A till sells without pause; after each power cut (plug pulled, or `--auto` hard reset) checks every confirmed sale whole (lines, payments = total, stock movements, audit entry), no half sales or orphan lines, stock consistent, both databases' integrity
`tools/gate/event-coverage.py`	Every record of every table has its create event (and changed ones an update event); read-only
`docs/gate/P1-gate.md`	The gate: each item, how, result; NFRs; how to run

2. Load tests on the Pi (5 tills, 10 minutes each)

Run	Sales	Memory while selling	Peak	Payment p50 / p95	Data checks
Stress (a sale every ~1.5 s per till), before the fix	600 + 30 offline, 69 returns, 25 voids	up to 132 MB	**244 MB** (end checks)	0.46 / 1.08 s	all pass
Stress, after the fix	541 + 30 offline	up to 148 MB	158 MB	0.48 / 1.19 s	all pass
Stress, 6 JS engines	539 + 30 offline	up to 121 MB	172 MB	0.51 / 1.43 s	all pass
Store pace (a sale every ~15 s per till)	156 + 30 offline	**levels off ~99 MB**	**138 MB**	0.47 / 1.13 s	all pass

Hub's own time for a sale (its request log, store pace): median 450 ms, p95 1,691 ms; quote median 181 ms. Wi-Fi + Caddy alone: median 175 ms, spikes to 2.2 s.

How the memory problem was found

The spike came after selling, while the test read thousands of records. Measured on the Pi, sampling 5 times a second:

stock movements 500 a page → 232 MB, 100 a page → 104 MB; sale lines (cost hook) 215 MB; audit log (no hook) 160 MB. On the laptop as a manager: 1,194 sale lines → 51 → 416 MB in 4.3 s; audit log 7,374 rows → 105 MB in 0.8 s. The cost-hiding hook loaded the whole access library and asked the database for the person's role and overrides for every record.

3. Fixes (DL-108)

File	Change
`hub/pb_hooks/lib/costs_hook.js`, `access.pb.js`	Small per-record hook; "may see costs" cached 10 s per person in the hub's store; registered also for **returns and return lines**
`hub/pb_hooks/lib/access.js`	App users: at most 200 rows a page of tables with costs (400 above)
`client/src/lib/api.js` (`apiAll`), `Stock.svelte`, `Counts.svelte`	Read page by page, 200 at a time (they also stopped at 1,000 products)
`hub/pb_hooks/lib/sales.js`, `stock.js` (`lotsOf`)	A sale reads each product's lots once (was up to 4 times) and each tax class once (was up to 4 times a line)
`hub/tests/step19-reports.test.mjs`	+3: big pages refused, cashier sees no return costs, manager does

After: laptop 1,194 sale lines → 273 MB (lean hook); a 200-row page → 152 MB; on the Pi a 100-row page 104 MB.

4. Power cut mid-sale (3 automatic hard resets)

Cycle	Confirmed sales	Lost	Incomplete	Half sales	Stock	Integrity	Back in
1	105	0	0	0	ok	ok / ok	38 s
2	208	0	0	0	ok	ok / ok	39 s
3	311	0	0	0	ok	ok / ok	40 s

No one had used the store's hub for over an hour before the resets. The store's data before/after: 4 sales, $73.13, 4 payments, 33 stock movements, 16 products: unchanged; integrity ok. Test hub removed.

5. Backup and audit log

Backup made and verified; restore test 0.5 s, 29 tables equal (4 sales, 16 products). Event coverage: store 338/338, load-test hub 5,104/5,104. Hub 46 MB, 51.5 °C.

Tests: **782 passed, 0 failed** (20 suites).

Failed cases and fixes

Symptom	Cause	Fix
Hub memory 244 MB in the first load test	Cost hook per record (library load + permission lookups) on 500-row pages	DL-108
Cashiers could read the cost of returned goods	Returns tables missing from the hook registration (step 6)	Registered; tests
Hub refused nothing above 200 at first, then crashed	`e.request` does not exist in list hooks	`e.requestInfo().query`
Load test: NFR-02 "failed" on the second local run	The dev hub still had the first run's sales	Count only this run's tills
Load test could not start with one till	No manager to receive stock	Manager first
Every sale failed after the lots change	A new `const lots` clashed with `let lots` in the same block	Renamed; sales and stock suites pass
Cases and 6-packs could not be sold in the test; perishables refused	Only loose stock received, no expiry	Receive each selling unit, expiry in 30 days
A temporary switch-off of the cost hook (to measure)	Declined by the permission system (security hook)	Measured against a table without the hook instead

6. Decisions and sprint plan (Sreya, 2026-10-08)

- **DL-107 decided:** the payment time on the Pi Zero is accepted for the pilot; the hub moves to a **Pi 4** later.
- **Hub memory question:** the problem is fixed (store pace ~99 MB, peak 138 MB, under the 150 MB target). The Pi Zero 2 W has 512 MB soldered on the board (415 MB usable after the graphics share; checked with `free -m`: 242 MB available, 14 MB of 414 MB compressed swap used). It cannot be upgraded; the hub's limits (MemoryMax 220 MB, GOMEMLIMIT 100 MiB) can be raised a little but that takes memory from the system. The Pi 4 (2-8 GB) is the real increase; its limits will be raised then.
- **Agile:** each phase is one sprint (`docs/sprints/README.md`). Sprint 1 = P1 Sell (`docs/sprints/Sprint-1.md`): build done; today Sreya runs the functional tests and gets the accountant review (Q1) by end of day.

For Sreya

1. Functional testing with `docs/testing/Chedam-Manual-Tests-P1-1-10.xlsx` (109 checks). Rows marked NEEDS PRINTER: Blocked for now.
2. The gate rows L01-L06 need you: 3 real devices selling together, the Android camera, 10 real power pulls (I run the tool, you pull the plug), the spare-card restore, a pilot week.
3. DL-107: decided (accepted for the pilot, Pi 4 later).
4. Still open: Q1 accountant review (blocks the P1 release), Q2 return defaults; replace the temporary GST/PST/BN numbers.

Next

1. Sreya's functional testing and the gate rows with real devices; fix what she finds.
2. P1 release after the gate and Q1.

2026-10-07 (Wednesday, October 7, 2026)

Fixes from Sreya's testing; P1 step 3 deployed; P1 step 4 offline selling built, tested and deployed; till opens offline; P1 step 5 receipts and printers built and deployed; P1 step 6 returns, exchanges and store credit built and deployed; P1 step 7 shelf labels built; manual test sheet; Pi health, backup and power-cut checks; P1 step 8 import and export built and deployed; P1 step 9 money and audit reports built

Summary

- **Sreya's report:** (1) the setup wizard showed "3 steps left" (Language, Clock, Owner account) but those steps could not be opened; (2) the laptop showed "Not a paired device" after pairing; (3) the sample PIN for Demo Owner did not work; (4) the camera scanner works on the iPhone (R5 ✓).
- **Cause (1):** those three steps are completed by the hub's first start; the dev store was made by sample data, so they were never marked, and the wizard had no screens for them. Fixed: screens for Language, Clock and Owner account.
- **Cause (2):** the Surface **was** paired at 11:54:54 (hub records), but no sign-in succeeded after it (one wrong PIN at 11:55:03, no later sign-in). The page in the screenshot was an owner session started earlier with the password on an unpaired browser. Fixed: home now says so and offers "Pair this browser".
- **Cause (3):** Demo Owner's PIN is no longer 1357 (checked offline against the stored hash on the Pi; the four other sample PINs still match). It was changed at some point (an admin-UI edit of Demo Owner happened at 11:45). The audit log could not show it because secret fields were left out entirely. Fixed: the log now names a changed secret field (`pin`, never its value). **Not reset**: Sreya signs in with the owner email and password on the paired laptop, then Owner account → Change my PIN.
- Step 3 (selling) and the fixes deployed (`dev-50b7ad0`), then step 4 (`dev-9a17683`).
- **P1 step 4, offline selling:** the till prices sales with the hub's own code, sells through an outage (also after a reload), prints offline receipts and uploads them when the hub is back. 584 tests pass.

1. Fixes (commit `50b7ad0`)

File	Change
`client/src/screens/wizard/Language.svelte`, `Clock.svelte`, `Owner.svelte`; `Wizard.svelte`	The three first-start steps can be opened, completed and re-run (FR-1.15). Clock compares hub and device time and sets the store time zone; Owner account edits name, email, password, links to Change my PIN and a new recovery code
`client/src/screens/Home.svelte`	"This browser is not paired" card with **Pair this browser**; Sell and Stock no longer listed as "coming in a later phase"
`hub/pb_hooks/lib/event_log.js`	Hidden fields that change (PIN, password, device key) are named in `changed`, never their values
Tests	step2: PIN changed in the admin UI is logged by name only; step4: no hash values in the log (names allowed), a revoked device shows `key_hash` changed

Checked in the browser on a fresh dev hub (same 3 steps open): Language → Clock ("This device and the hub agree") → Owner account → Business profile; 3 of 10 done.

Deploy: `bash hub/scripts/deploy.sh --sample-data` → `dev-50b7ad0`, integrity ok, 35 permissions, sales/till settings present, 53 MB.

2. P1 step 4: offline selling

File	What

`hub/pb_migrations/1791300008_p1_offline.js`	`sales.offline_ref`, `synced_at`, `sync_note`
`hub/pb_hooks/lib/offline.js`	Offline pack; upload of an offline sale: arithmetic checked with `pricing_core.js`, recorded as charged, S- number, stock may go negative
`hub/pb_hooks/lib/stock.js`	Urgent "oversold while offline" task per product, closes itself when stock is back at 0 or more
`hub/pb_hooks/sales.pb.js`	`GET /api/chedam/sales/offline-pack`, `POST /api/chedam/sales/offline`
`client/vite.config.js`	`virtual:pricing-core`: the app imports the hub's pricing file at build time
`client/src/lib/offline_price.js`	Offline lookup and quote, same rules as the hub
`client/src/lib/offline.svelte.js`	Dexie (IndexedDB): pack and queue; upload in order every 10 s
`Sell`, `Till`, `Receipt`, `ConnectivityBar`, `App`, `session`, `api`	Offline prices, offline receipts, waiting count, till close waits for uploads, signed-in person kept over an offline reload
`hub/tests/step14-offline.test.mjs`	27 tests, including **parity**: the till's offline prices equal the hub's quote on 6 carts (basket, discounts, override, exemption, packs, weights)
`tools/devhub/dev-login.mjs`	Pairs and signs in a sample person on the local dev hub (for browser checks)

Decisions DL-86..89 in `docs/P1-mini-spec.md`.

Checked in the browser (local dev hub, Cal Cashier)

1. Opened the till; Sell downloaded the pack (11 products, in IndexedDB).
2. **Stopped the hub.** Cola ×2, chips, bananas 1.235 kg → "offline prices", **$9.88** / **cash $9.90** (same as online).
3. First payment try: "This device could not save the sale … do not hand over the goods" (safety message worked) → bug found and fixed (below).
4. Started the hub, reloaded, **stopped it again and reloaded during the outage**: the till came back to Sell as Cal Cashier with the cart (bug found and fixed below).
5. Paid $20 → **OFF-X3G8-0002**, change $10.10, full receipt, "1 offline sale waiting".
6. **Started the hub:** the queue emptied by itself; on the hub: **S-000001**, offline ref OFF-X3G8-0002, total 9.88, rounding 2 ¢, change 10.10, linked to the till, 3 sale movements "S-000001 (offline OFF-X3G8-0002)".

Deploy: `dev-9a17683`, migration `1791300008` applied, no errors, integrity ok, 49 MB.

3. Sreya's offline test and the change it led to (DL-90)

Sreya's test on the Pi (`dev-9a17683`):

- **Case 1:** Pi on, products in the cart, Pi switched off before payment → the sale completed and the receipt printed. ✓
- **Case 2:** Pi switched off **before** the till was opened → the till could not be opened, so nothing could be sold. The Till screen only said "The hub is not answering".
- Sale and till close (Z report) checked online. ✓

Cause: opening a till needed the hub (by design in step 4). In a store this means no selling at all if the Pi or Wi-Fi is down at opening time. Sreya chose to allow opening offline.

File	Change
`hub/pb_migrations/1791300009_p1_offline_tills.js`	`tills.offline`, `device_time`, `synced_at`, `sync_note`

`hub/pb_hooks/lib/tills.js`, `sales.pb.js`	`POST /api/chedam/tills/offline-open`: the till's own id (repeat = first result), float, opener, the device's opening time if within the last 24 h; numbered on arrival; another till still open on the device is noted with a task
`hub/pb_hooks/lib/offline.js`	Offline pack also carries the default float and the denominations
`client/src/lib/offline.svelte.js`	Till opening queued ahead of the sales; the device's till kept in one place (`tillInfo`)
`Till.svelte`	Offline: explains what works, opens the till on the device; cash drops, pay-outs and closing wait for the hub
`Sell.svelte`, `ConnectivityBar.svelte`	"Till open (opened offline)"; waiting count includes the till opening; online, the till opening uploads before the till is shown
`hub/tests/step14-offline.test.mjs`	8 tests: number, float, repeat, other device refused, sale lands in it, Z figures, second open till noted, opening time
`tools/devhub/dev-login.mjs`	Missing `readFileSync` import (the helper did not run)

Checked in the browser (local dev hub, Cal Cashier): signed in, **stopped the hub**, Till → counted $100 (5 × $20) → "Till (opened offline)", sold a cola for $1.75 cash → OFF-P9X3-0001. **Started the hub:** the opening and the sale uploaded by themselves; Till screen showed **Till 1**, 1 sale, expected cash **$101.75**.

Tests: **592 passed, 0 failed**. Not deployed yet (waiting for Sreya's go-ahead).

4. Deploy of the offline till opening

Sreya's go-ahead → `bash hub/scripts/deploy.sh --sample-data` → **`dev-7a51b25`**, migration `1791300009` applied, API healthy, no errors.

5. P1 step 5: receipts, printers and the cash drawer (no hardware yet)

Sreya has no receipt printer or drawer yet, so everything was built against the standard ESC/POS commands and checked with a pretend printer that shows exactly what a real one would receive.

File	What
`hub/pb_hooks/lib/receipt_layout.js`	Receipt, Z/X report and test page as one layout → ESC/POS bytes, or the same lines as text; 48/42/32 characters; accents simplified for the printer
`hub/pb_hooks/lib/printing.js`, `bin/printer.sh`, `printing.pb.js`	Printers setting and per-device printer, network scan (port 9100), test print, print/reprint a sale, receipt as text, Z/X report, drawer for a no-sale
`hub/pb_migrations/1791300010_p1_printing.js`	`sales.reprints`, `sales.drawer_opened`, `cash_movements.drawer_opened`; settings `printing.printers`, `printing.receipt`
`client/src/screens/Printers.svelte` (Home → **Receipt printer**)	Find printers, add by address, test print (with drawer), paper width, which device prints where, print when paid, full-receipt threshold, preview of the last sale as printed
`PrintButtons.svelte`, `printer.svelte.js`; `Sell`, `Sales`, `Till`, `Receipt`	Receipt prints when paid (drawer opens for cash), Reprint (COPY), print on this device, customer's name for $150+; no-sale opens the drawer; Print X / Z report; on-screen receipt shows Total → Cash rounding → Total in cash
`hub/tests/step15-printing.test.mjs`	31 tests with a fake network printer
`tools/devhub/fake-printer.mjs`	Pretend printer for the laptop: shows each job as text, marks [DRAWER OPENS], [BARCODE], [CUT]

Decisions DL-91..93 in `docs/P1-mini-spec.md`.

Checked in the browser (local dev hub, Mira Manager, fake printer on 127.0.0.1:9100)

1. Receipt printer → Add by address → **Test print first**: test page arrived (48-character ruler, bold total, barcode, cut). Saved.

2. Opened the till, sold a cola and chips, paid $10 cash → "Printed on Receipt printer · drawer opened"; the printer got [DRAWER OPENS], the receipt with till number, GST/PST lines, change $4.05, [BARCODE S-000001], [CUT].
3. **Reprint** → "copy 1", printed " ***COPY (reprint 1)*** ", no drawer.
4. Till → No sale with a reason → "No-sale recorded. Drawer opened." (kick only). **Print X report** → printed with expected cash $205.95.
5. Preview at 58 mm: every line within 32 characters.

Tests: **623 passed, 0 failed** (16 suites). Not deployed yet (waiting for Sreya's go-ahead).

6. Deploy of step 5

Sreya's go-ahead → **`dev-8820126`**, migration `1791300010` applied, API healthy, no errors. On the Pi, the hub's own user ran the printer network scan: 9 s, no printer found (none connected), memory fine.

7. P1 step 6: returns, exchanges and store credit

File	What
`hub/pb_migrations/1791300011_p1_returns.js`	`returns`, `return_lines`, `store_credits`, `refunds`; payment method "exchange"; permission `sales.return`; return policy settings (Q2 defaults)
`hub/pb_hooks/lib/returns.js`, `returns.pb.js`	Find the sale; what can still come back; refund per line (BR-23); approvals (BR-17); no receipt at the lowest 30-day price as store credit; stock per item; refunds; exchange in one transaction
`hub/pb_hooks/lib/sales.js`	Store credit as payment (balance checked and reduced; given back on void); exchange payment only from a return; sales with returns cannot be voided
`hub/pb_hooks/lib/tills.js`, `receipt_layout.js`, `printing.js`	Z/X report: returns, tax refunded by type, refunds by method, cash refunds out of expected cash; return slip with store credit barcode and signature line
`hub/pb_migrations_dev/1791300102_dev_sample_numbers.js`	**Temporary** GST/HST `999999999 RT0001`, PST `PST-9999-9999`, BN `999999999` in the sample store (fills empty fields only)
`client/src/screens/Returns.svelte` (Sell → **Return**)	Find by scan/number, date, amount, card last 4, or no receipt; quantities and where each item goes; refund and PIN; refund chooser; slip printed
`Sell.svelte`, `RefundChooser`, `ReturnSlip`, `exchange.svelte.js`, `PrintButtons`, `Till.svelte`	Exchange finished on Sell (difference paid or refunded); Store credit payment; return slips printed (drawer for cash refunds); returns on the Till screen
`hub/tests/step16-returns.test.mjs`	43 tests

Decisions DL-94..97 in `docs/P1-mini-spec.md`.

Checked in the browser (local dev hub, Mira Manager, fake printer)

1. Return → scan/type **S-000001** → 1 of 2 chips → refund $3.99 + GST $0.20 = **$4.19** → Cash **$4.20** (rounded) → **R-000001**; slip printed with [DRAWER OPENS], the temporary GST/PST numbers, signature line, [BARCODE R-000001].
2. Exchange: 1 cola back ($1.76) → Sell → chips $4.19 → pay screen showed exchange credit $1.76, to pay **$2.43**, cash $2.45 → paid $5, change $2.55 → **S-000002 + R-000002** recorded together, both printed.
3. Till screen: Returns 2 · $5.95 (cash $4.20); expected cash $200 + $16.15 − $4.20 = **$211.95**.
4. No receipt: scanned a chocolate bar → only "Store credit $3.13" offered → **SC-…** code printed.

Tests: **666 passed, 0 failed** (17 suites).

8. Deploy of step 6

Sreya's go-ahead → **`dev-63986da`**, migrations `1791300011` and `1791300102` applied, API healthy, no errors. On the Pi: 36 permissions; business numbers now the temporary `999999999 RT0001` / `PST-9999-9999` / `999999999`.

9. P1 step 7: shelf labels

Sreya will test steps 5, 6 and 7 together.

File	What
`hub/pb_migrations/1791300012_p1_labels.js`	`label_layouts` (6 presets), `label_templates` (Shelf label, Bilingual shelf label), `label_batches`, `label_batch_items`; product size (`size_qty`, `size_unit`); permission `labels.manage` (manager, staff)
`hub/pb_hooks/lib/labels.js`, `labels.pb.js`; `lib/catalogue.js`	Batch with merging; automatic adds on a new price or a product going on sale (BR-25, Labels module on); add by code, category, list; batches as snapshots; "printed" with the changed-since check; last 10; layout fit check
`client/src/lib/labels/`	`barcode.js` (EAN-13/UPC-A/EAN-8/CODE128), `unitprice.js`, `geometry.js`, `pdf.js` (jsPDF, exact size, alignment page)
`client/src/screens/Labels.svelte` (Home → **Labels**)	To print · Printed · Layouts · Templates; start position picked on a drawing of the sheet
`ProductEdit.svelte`	"Size of one" (for the unit price)
`client/package.json`	`jspdf` 4.2.1 (exact; 0 vulnerabilities; loaded only on the Labels screen)
`hub/pb_migrations_dev/1791300103_dev_sample_sizes.js`	Sizes for the sample products (only where empty)
`hub/tests/step17-labels.test.mjs`	38 tests

Decisions DL-98..100 in `docs/P1-mini-spec.md`.

Checks

1. Barcodes: every code decoded back by ZXing (the scanner library), all 95 CODE128 characters; then a sample sheet rendered at 300 dpi and read row by row: all barcodes decoded, including a UPC-A 6-pack.
2. Labels looked at as images (A4 3×8 and Letter 3×10 with cut lines): name, French name, size, origin, price /kg or pack name, unit price, barcode or PLU; nothing overlapping.
3. Browser (local dev hub, Mira Manager): Labels showed the 14 sample products waiting ("new product"); start position 4 picked on the sheet drawing → **Make the PDF** → real PDF, 14 labels from position 4 → **Printed fine** → batch empty, "Batch 1 · 14 labels" under Printed. Layout editor: 4 columns on A4 → "Too wide: the labels need 268.7 mm of 210 mm", Save disabled; alignment page opened.

Tests: **704 passed, 0 failed** (18 suites).

10. Manual test sheet, Pi health, backup and power-cut test

Manual test sheet for Sreya: `docs/testing/Chedam-Manual-Tests-P1-1-7.xlsx` (85 checks in 9 areas: setup and access, products, stock, selling, offline, receipts and printer, returns, labels, reliability; Pass / Fail / Blocked / Not tested; Summary tab). Built by `tools/testing/build_test_sheet.py`.

Pi health (read-only), 11:39: hub active, 0 restarts, version `dev-63986da`; 57 MB hub memory (target < 150 MB), 235 MB free; 44 °C, no throttling or under-voltage; disk 3% used; clock synced (America/Vancouver); database integrity ok (WAL); no hub errors in 24 h; no SD card errors; Wi-Fi guard online; HTTPS `chedam.local` 200 in 0.2 s from the laptop; site certificates renew every 12 h, root valid to 2036; data 8.5 MB.

Backups: drive = the 7.4 GB USB stick (not the Seagate). Nightly backups verified 2026-10-06 and 2026-10-07 (02:30). Manual backup now: 174 KB encrypted, verified on the drive in 1.7 s; restore test 0.4 s, counts equal. - **Gap found and fixed:** the restore test compared only the P0 setup tables. `chedam-backup` now also compares products, stock, tills,

sales, payments, returns, store credit and labels. Run on the Pi from a temporary copy: all 27 tables equal to live (sales 3 = 3, sale lines 8 = 8, payments 3 = 3, stock movements 20 = 20, …). Installed on the Pi with the next deploy.

Power-cut (data loss) test: the P0 power test hub (throwaway database on the same SD card, `tools/powertest/`) with writes every ~70 ms from the laptop; instead of pulling the plug, the Pi was hard-reset with `sysrq b` (immediate reboot, no sync, no unmount) 40 s into each cycle.

Cycle	Confirmed writes	Lost	Duplicates	Audit log	Integrity (store / test)	Store hub	Back in
1	972	0	0	matches	ok / ok	active	40 s
2	1976	0	0	matches	ok / ok	active	43 s
3	3017	0	0	matches	ok / ok	active	42 s

The store's own data before/after: 3 sales, 5,633 cents, 3 payments, 20 stock movements: unchanged; integrity ok. A product Sreya saved 10 s before the third reset ("Usb2.0 otg") was kept. Report: `docs/gate/powertest-2026-10-07-p1-sysrq.json`. Test hub removed. A real plug pull stays in the manual sheet (I01, I02): sysrq leaves the SD card powered.

11. Deploy of step 7 and the backup check

Sreya's go-ahead → **`dev-bc4c3e5`**: migrations `1791300012` (labels) and `1791300103` (sample sizes) applied, backup helper updated, API healthy, no errors. On the Pi: 6 layouts, 2 templates, 37 permissions, 8 sample products with a size, 0 labels waiting (existing products: labels appear on the next price change, or add them by category). Restore test with the installed helper: consistent, 27 tables compared. Database integrity ok.

12. P1 step 8: product import and data export

File	What
`client/src/lib/import/read.js`, `profile.js`, `map.js`, `transform.js`	Read CSV/TSV/text, Excel, JSON, Parquet, zip; encoding, delimiter, header row; column profiles; mapping with confidence; transforms
`client/src/screens/ImportExport.svelte` (Home → **Import and export**)	File → Columns → Values (categories, tax, options) → Check (filters, fix a row, set a value for the rows shown, exclude) → Import; job log; full export for the owner
`hub/pb_hooks/lib/imports.js`, `imports.pb.js`	Check and all-or-nothing import with a job record
`hub/pb_hooks/lib/catalogue_save.js`	The product form's save, shared with the import (same rules)
`hub/pb_hooks/lib/data_export.js`	Data dictionary and tables page by page, without secrets
`client/src/lib/export.js`, `components/ExportMenu.svelte`	Export on Products, Stock and Sales (CSV, Excel, PDF); full export zip
`hub/pb_migrations/1791300013_p1_import_export.js`	`import_jobs`; permission `data.export` (owner only)
`client/package.json`	papaparse 5.7.0, SheetJS 0.20.3 (vendor channel), hyparquet 1.31.2, fflate 0.8.3 (exact; 0 vulnerabilities; loaded only on the Import screen)
`hub/tests/step18-import-export.test.mjs`	45 tests

Decisions DL-101..103 in `docs/P1-mini-spec.md`. DL-101 changes DL-03 for imports (no DuckDB): see the reason there.

Checked in the browser (local dev hub, Mira Manager)

A Shopify-style CSV (5 rows): all columns mapped with high confidence; Beverages matched, Pantry to be created; "GST+PST" → Standard, "Zero" → Zero-rated; check: duplicate barcode on line 5 caught, Cola found as already there, cost above price and the 6×355 mL pack noted; line 5 excluded → **3 new, 1 left as it was, 1 excluded, 1 category created, 0.3 s**, logged. Products → Export → CSV, Excel and PDF made.

Tests: **749 passed, 0 failed** (19 suites; the rate-limit suite once timed out starting while the laptop was busy, then passed alone). Not deployed yet (waiting for Sreya's go-ahead).

13. Deploy of step 8

Sreya's go-ahead → **`dev-495b5a6`**, migration `1791300013` applied, API healthy, no errors; 38 permissions, integrity ok, app 2.4 MB.

14. P1 step 9: money and audit reports

File	What
`hub/pb_migrations/1791300014_p1_reports.js`	Tills: card settlement, batch number, reconciled at/by, note; settings `retention.years` (6), `reports.flags`
`hub/pb_hooks/lib/reports.js`	Till reconciliation, reconcile, loss prevention per cashier with flags, the events behind them
`hub/pb_hooks/lib/audit.js`	Audit log search and filter choices; costs removed without costs.view; people by name
`hub/pb_hooks/lib/retention.js`, `reports.pb.js`	Hard deletes refused inside the window (also superusers); report endpoints
`client/src/screens/Reports.svelte` (Home → **Reports**)	Tills · Loss prevention · Audit log · Retention; every table exportable
`hub/tests/step19-reports.test.mjs`	30 tests

Decisions DL-104..106 in `docs/P1-mini-spec.md`.

Checked in the browser (local dev hub, Mira Manager)

A till with 3 sales (one by card) and a no-sale, closed $3 short: Tills showed "$3 short" and "Card settlement not entered"; Reconcile with batch B-1007 and no note → "Cash or card do not balance: add a note"; with a note → reconciled, terminal $3.54 = cards, 0 left to reconcile. Loss prevention: the manager's row and the store totals. Audit log, filtered to tills: the reconcile change with every field before → after. Retention: 6 years, the record counts.

Tests: **780 passed, 0 failed** (20 suites). Not deployed yet.

Failed cases and fixes

Symptom	Cause	Fix
Wizard steps Language/Clock/Owner greyed out	Marked done only by the hub's first start; no screens	Screens for the three steps
"Not a paired device" after pairing	The page shown was an owner password session from before pairing; pairing needs a new sign-in	Home explains it, "Pair this browser"
Sample owner PIN rejected	PIN changed since (checked against the hash on the Pi, no sign-in attempt used); the log did not show secret changes	Log names changed secret fields; Sreya changes her PIN after a password sign-in
Pack: barcodes came as `[91,93]`	JSON fields are raw bytes in the hub's JS engine, also through `publicExport()`	Parse `getString()` for JSON fields
Parity test "failed" with equal values	Key order differs between hub and till objects	Compare with sorted keys
Offline Z count expected 4, got 3	My test: 2 of the 5 uploads were refused on purpose	Expect 3
A stray `cat >` waited for input and the test ran 400 s	My command	Stopped; commands now read from `/dev/null`
Offline sale not saved ("storage is full or blocked")	IndexedDB cannot store Svelte's reactive proxies	Store plain copies
Reload during an outage went to the name list	Who is signed in was only in memory	Kept on the device, used only with the same person's unexpired token (DL-89)
Header said "Not a paired device" offline	Device details come from the hub	Uses the saved device key
A NUL byte and line breaks landed in `vite.config.js`	Escapes inside a here-document	Rewrote the plugin; checked no NUL remains

"1 sales waiting"	Plural	Fixed
Till could not be opened with the hub off (Sreya, case 2)	Opening needed the hub	Opens on the device, uploads first (DL-90)
Till opened offline showed the upload time	Hub used the arrival time	Uses the device's time when plausible
"0 sales and the till opening waiting"	Wording	Only what is waiting is named
Printing tests counted 2 jobs too many	The network scan's knock (an empty connection) looked like a print job	Fake printer ignores empty connections
Card sale in the test refused	My test paid a made-up amount	Pays the quoted total
"Deposit/fee" lost its indent under the item	Line wrapping dropped leading spaces	Indent kept on every wrapped line (test added)
X report showed "-$0.00"	Minus sign on zero	"$0.00" when nothing was taken off
Network scan finds nothing on the laptop	It reads the Pi's network (`ip`), which Windows lacks	Expected; on the laptop add the printer by address
No-receipt credit came out as $0	A product's first price is stored as a change from $0, and the lowest-price rule took the 0	Zero is never a selling price: ignored (test added)
Test used a price the lookup does not return	My test	Reads the unit's price
Schema test expected 35 permissions	Step 6 adds `sales.return`	36
EAN guard bars ran through the digits	Long guard bars under centred digits	Plain bars, digits under them
Price overlapped the barcode on a 6-pack label	Fixed half-width for the barcode	Barcode gets the space the price leaves; the price shrinks before the barcode gets too narrow to scan
"Produit de Ecuador"	French country names need their article	"Origine : Équateur" (and "Produit du Canada")
Route helper not found in a handler	Each handler runs in its own JS engine	Helper inlined (known rule, conventions.md)
Page size test: 612.0000000000001 pt	Rounding of 215.9 mm	Compared as numbers
Undoing the sample catalogue failed	Its products now have label lines	The sample's undo removes them first
Schema test expected 36 permissions	Step 7 adds `labels.manage`	37
A staff import would have failed outright	Staff may edit the catalogue but not prices; the import tried to activate	Staff imports are Drafts; existing prices left alone, with a warning
SheetJS landed in the till's main bundle	The import screen imported the file readers directly	Loaded with the first file
An unnamed column of 4-digit codes was mapped to HS code	A weak name guess beat what the values say	Values outrank weak names; PLU-like values never go to HS code
"Zero" tax was guessed as Exempt	First class matching "zero or exempt"	Zero → Zero-rated, Exempt → Exempt
"Category will be created" made rows Drafts	Shown as a Draft reason	A note instead
Export file names in UTC	ISO time	Local time
Schema test expected 37 permissions	Step 8 adds `data.export`	38
Reconciling with a terminal amount but no batch number skipped the card check	"Settled" was read from the batch number	Any reconcile counts as settled
Audit-log delete said "Audit log (never deleted) are kept for 6 years"	Generic retention message	"The audit log is never deleted."
Test: price override refused; manager's sales refused	My test: 12% override is over the 10% limit; the sample cola ran out	7% override; the test receives stock first

A lenient test check (late offline sale)	Accepted a failed upload	Made strict; passes
Audit log showed "reconciled_by: "	Person fields shown raw	Shown by name
Loss prevention total $0.02 off the till's	Cash rounding left out	Included

Tests: **584 passed, 0 failed** (15 suites).

For Sreya

1. **Sign in again on the laptop:** on the name list choose **Owner: sign in with email and password** (your own password), then Store setup → **Owner account** → **Change my PIN**. Use the same address every time on a device (e.g. always `https://chedam.local`): pairing is remembered per address.
2. Finish the 3 setup steps (Language, Clock, Owner account) and enter the **GST/PST numbers** in Business profile (they print on receipts).
3. Try **Sell**: open the till (Till → Open till), sell, close with the Z report. To test offline: switch the Pi off (or its Wi-Fi) during a sale, keep selling, switch it back on and watch "waiting to upload" go to zero.

Next

1. Sreya tests steps 5, 6 and 7 together with real cases; replace the temporary GST/PST/BN numbers before selling for real.
2. When a printer arrives: Find printers, test print, test with drawer, one cash sale. Labels: print the alignment page on plain paper first.
3. Deploy step 9 (needs Sreya's go-ahead). Then P1 step 10: the gate (3 devices selling, power pulls mid-sale ×10, offline queue, camera, memory under load, backup + restore).
4. Open (to plan in the coming days): Q1 accountant review (tax classes, deposits, exemption reasons), Q2 return defaults.

2026-10-06 (Tuesday, October 6, 2026)

P0 gate: power pull ×10 passed after two real fixes (Wi-Fi lost at boot, confirmed writes lost); untrusted-certificate warning

Summary

- **New Android phone** ("1plus", Android 10, Chrome 151) worked without the certificate installed. Not a bug: Chrome lets a person tap through "Your connection is not private". The app now warns with an amber bar (DL-62).
- **Power-pull gate test (NFR-09, ×10): PASS on run 3.** 47,573 confirmed writes, **0 lost**, 0 duplicates, audit log complete, both databases intact, the real hub restarted by itself every time (48-78 s).
- It found two real problems first, both fixed and proven: 1. **Wi-Fi lost after power cuts during boot**: the Wi-Fi settings files were left empty. Fix: `chedam-netguard` (DL-60), tested by deleting the profile: recovered unattended in 127 s. 2. **Confirmed writes lost in a power cut** (23 of 1,646): SQLite confirmed before writing to the card. Fix: `synchronous=FULL` (DL-61).
- My test script crashed once (too short a timeout after a cold boot); fixed, and confirmed writes are now kept in a file as they happen.
- Gate status: 3-device login ✓, update + rollback ✓, backup + restore ✓, memory under load ✓, power pull ×10 ✓. Left: camera over HTTPS.

1. New phone works without the certificate

Hub record	"1plus", approved, Android 10, Chrome 151, app 0.9.0
Why it worked	Chrome's warning page "Your connection is not private" has Advanced → **Proceed (unsafe)**. After that the page works, but the hub is not verified (an impostor on the Wi-Fi would not be noticed) and Chrome refuses the app's offline mode (service worker) and may refuse the camera
Fix	The app detects the service worker's certificate error and shows an amber bar: "This device does not trust the hub's certificate yet, so offline mode and the camera will not work. **Install the certificate**" (link to the HTTP guide). Commit `61d86f0`, deployed

2. Power-pull test tools

File	Role
`tools/powertest/pi-powertest.sh`	`install`: a throwaway test hub that survives reboots (systemd service `chedam-powertest`, same binary, hooks, migrations and memory limits as the real hub, data in `/var/lib/chedam-powertest` on the same SD card, 127.0.0.1:8199); `check`: integrity of both databases, services, records vs. audit entries, power flag, boot id; `remove`
`tools/powertest/power-writer.mjs`	On the laptop through an SSH tunnel it restarts by itself: 3 writers (about 26 writes per second, each with its audit entry), notices the cut and the **reboot** (boot id), then checks: every confirmed write present, no duplicates, audit = records, integrity, real hub active

Dry run with a software reboot: 520 confirmed writes, 0 lost, PASS.

3. Run 1: the writer crashed, then the Pi came back without Wi-Fi

Time (UTC)	What happened
16:04	"Hub gone" for 6 s, no reboot: the laptop's SSH tunnel dropped (Wi-Fi blip); correctly not counted. Sreya had not pulled yet
16:06	Pull 1 → back after a reboot at 16:07

16:07	**My script crashed** while checking: listing ~17,000 records on a freshly booted Pi Zero took more than its 4 s timeout. The list of confirmed writes was only in memory, so this cycle could not be judged
after	Sreya pulled two more times; the Pi stopped answering (ping, SSH)

On the hub's monitor: a normal login prompt, but "My IP address is **127.0.1.1**" (no network address). The router listed 3 Wi-Fi devices, none the Pi. Sreya connected a keyboard:

Check	Result
`nmcli dev status`	`wlan0 wifi disconnected`
`rfkill list`	not blocked
`dmesg`	Wi-Fi firmware loaded normally (BCM43430/1)
`nmcli con show`	**only `lo`**: no Wi-Fi network known at all

Sreya reconnected with `sudo nmcli dev wifi connect "TP-LINK_2CBE" password "…" ifname wlan0` (her password, typed by her), and the hub was back at 192.168.50.101.

Root cause

- Raspberry Pi OS 13 keeps the Imager Wi-Fi settings in `/etc/netplan/90-NM-*.yaml`, from which NetworkManager's profile is generated **at every boot**.
- Both files were **0 bytes**: they are rewritten during boot, and a power cut before the data reached the card left them empty (the new length was saved, the contents were not).
- Sreya's new profile carried yesterday's time (04:49), because the Pi's clock was wrong at boot: no network to set it and no RTC (the DL-23 risk, seen live).
- Data after these three cuts: both databases intact; 17,863 test records = 17,863 audit entries.
- My setup script only adds a power-save setting; it does not touch the Wi-Fi profiles.

Fix: chedam-netguard (DL-60)

- `hub/system/chedam-netguard` + `.service` + `.timer` (every minute), installed by `deploy.sh` and `setup-hub.sh`:
- **online**: an fsync'd spare copy of each valid Wi-Fi profile (must have an SSID and not be empty);
- **2 min without an address**: restore the spare copies if no Wi-Fi network is known, then reconnect;
- **10 min**: reboot, at most once an hour.
- **Controlled test on the Pi:** the profile was removed and Wi-Fi disconnected (with a 9-minute safety timer that would have put it back). Log:
- 09:43:55 profile removed;
- 09:46:33 "no Wi-Fi network known after 127 s … restored spare copies";
- 09:46:37 "reconnected wlan0".

The hub was back at 192.168.50.101 with nobody touching it. The safety net was not needed (removed).

Fix: test script

- Checks after a reboot get 60 s with 3 attempts, and a failed check is reported, not a crash.
- Every confirmed write is appended to a file at once.

4. Run 2: confirmed writes lost (stopped after cycle 1)

Check	Result
Integrity, duplicates, audit, hub restart	all fine
Confirmed writes lost	**23 of 1,646**

- **Cause:** PocketBase opens SQLite in WAL mode with `synchronous=NORMAL`, which confirms a commit before forcing it onto the card. In a power cut the last confirmed transactions can vanish. That breaks NFR-02 ("no lost or

duplicated sales").

- **Fix (DL-61):** `hub/pb_hooks/durability.pb.js` sets `PRAGMA synchronous = FULL` on PocketBase's single non-concurrent (write) connection before every request, at start-up and every minute.
- Health page item "Saved before confirming" (bad if not FULL); test added; **329** tests pass.
- On the Pi: "Saved before confirming: Yes"; 30 writes in 2,147 ms (about 70 ms each, including the audit entry).
- Report kept: `docs/gate/powertest-2026-10-06-run2-synchronous-normal.json`.

5. Run 3: PASS 10/10

Cycle	Confirmed so far	Lost	Duplicates	Audit = records	Integrity real / test	Hub back after
1	873	0	0	yes	ok / ok	49 s
2	7,746	0	0	yes	ok / ok	48 s
3	10,239	0	0	yes	ok / ok	53 s
4	10,530	0	0	yes	ok / ok	51 s
5	10,775	0	0	yes	ok / ok	54 s
6	11,399	0	0	yes	ok / ok	64 s
7	11,770	0	0	yes	ok / ok	54 s
8	13,123	0	0	yes	ok / ok	65 s
9	13,845	0	0	yes	ok / ok	61 s
10	47,573	0	0	yes	ok / ok	78 s

- Each boot logged "EXT4 orphan cleanup", the normal filesystem repair after a power cut.
- Wi-Fi came back on its own every time (netguard did not need to act).
- Power was clean (`throttled=0x0`).
- 1-2 writes per cut were saved without being confirmed: they reached the card, but the answer was lost with the power. That is expected; the till retries them (idempotent ids in later phases).
- Report kept: `docs/gate/powertest-2026-10-06-run3.json`.
- The test hub was removed; the real hub is active.

6. Session 2: status check and a race fix

- Git: clean, `main` in sync with GitHub before starting. The Pi runs `dev-484c657`; the later commit `aebd3d5` was documents only.
- Full test run (`PB_BIN=C:/Tools/pocketbase.exe node hub/tests/run-all.mjs`): **1 failure**, `step9-updates` "one urgent task". It passed 3 times when run alone, so it was intermittent.
- **Cause (a real bug, not only the test):** when the hub applies an install result, it saved the update record and then its urgent task as two separate writes. The every-minute job and the status screen can both apply the same result at the same moment, so a reader could see "rolled back" before the task existed, and two appliers could each raise a task. `lib/backup.js` had the same pattern.
- **Fix (commit `85da79a`):** in `hub/pb_hooks/lib/updates.js` and `hub/pb_hooks/lib/backup.js`, finding the record, saving it and raising/closing the task now run in one transaction. The hub has a single writer, so the second applier waits, then finds nothing left to apply.
- **Verified:** full run 329/329 passed; `step9-updates` and `step7-backups` then passed 5 times in a row each.
- Not deployed yet: needs Sreya's go-ahead.

7. Session 2 (continued): P0 closed, P1 started

P0 closed (DL-63)

- Sreya: skip the camera test; the certificate trust step can be solved later.
- Recorded as DL-63 in `docs/P0-mini-spec.md`: camera over HTTPS (R5) and the phone trust step move to P1

step 2 (phone scanning) and are tested before the P1 gate. The pilot week waits for a pilot store (Q9); P0 and P1 are piloted together.

P1 plan

- `docs/P1-mini-spec.md`: goal, 10 build steps with requirement IDs, decisions DL-64..71, open questions (Q1 accountant blocks the P1 release, not the build; Q2; Q7; who sees costs).

P1 step 1, hub side: catalogue and tax tables (commit `d078aad`)

File	What
`hub/pb_migrations/1791300001_p1_catalogue_schema.js`	`tax_types`, `tax_rates` (effective dates), `tax_classes`, `deposits_fees`, `categories`, `products`, `selling_units` (single, pack, case, by weight; nested), `price_history`
`hub/pb_migrations/1791300002_p1_catalogue_reference.js`	Permissions `catalogue.edit` (staff, manager), `prices.edit` and `tax.manage` (manager); GST 5%, BC PST 7%; classes Standard, GST only, Zero-rated, Exempt; all marked pending review (Q1)
`hub/pb_migrations_dev/1791300100_dev_sample_catalogue.js`	8 categories, 12 sample products (bananas by kg, cola single/12-pack/case, water single and 6-pack sharing a barcode, an age-restricted item, one Draft)
`hub/pb_hooks/lib/catalogue.js`	BR-07 checks (what keeps a product a Draft), base quantity of nested packs, barcode and PLU uniqueness, price history, the Drafts task, lookup and product view
`hub/pb_hooks/lib/tax.js`	Tax rates for a class at a date; checks on rates and classes
`hub/pb_hooks/catalogue.pb.js`	`POST /api/chedam/catalogue/products` (product + units in one transaction), `GET /api/chedam/catalogue/lookup?code=`, `GET /api/chedam/catalogue/products/{id}`
`hub/pb_hooks/lib/event_log.js`	Table rules run inside the save's transaction (`SIDE_EFFECTS`); "before" read from the database
`hub/pb_hooks/lib/access.js`	New tables; price changes and activation need `prices.edit`
`hub/tests/step11-catalogue.test.mjs`	67 tests

Command: `PB_BIN=C:/Tools/pocketbase.exe node hub/tests/run-all.mjs` → **396 passed, 0 failed** (11 suites). Not deployed to the Pi yet (needs Sreya's go-ahead). Product screens are next.

Sreya's answers and the deploy

- Deploy: approved. Accountant review (Q1): in progress. Cashiers: no costs or margins.
- **DL-72 (commit `f75d9a1`):** new permission `costs.view` (owner, manager, accountant, staff). Without it the hub hides `cost_cents` on products and the amounts of cost-history rows in every API answer (lists, single records, realtime, through `onRecordEnrich` in `access.pb.js`), and the product view endpoint leaves them out. Migration `1791300003_p1_costs_view.js`; 7 tests. Full run: **403 passed, 0 failed**.
- **Deployed** with `bash hub/scripts/deploy.sh --sample-data`: pre-deploy backup integrity ok, version `dev-f75d9a1`, migrations `1791300001..3` and `1791300100` applied, no errors.
- **Checked on the Pi** (`sqlite3` as `chedam-hub`): 11 active products and 1 Draft; "Case of 2 x 12" base quantity 24; GST 5% and BC PST 7%, both pending review; Drafts task open ("1 product is a Draft and cannot be sold"); 26 price-history rows, each with its event; integrity ok; PocketBase memory 54 MB; health 200 over HTTPS from the PC; the lookup refuses a request without sign-in (401).

Failed cases and fixes (session 2)

Symptom	Cause	Fix
Sample data migration failed: "base unit cannot change"	PocketBase's `record.original()` still shows the pre-create values when the same record object is saved a second time	Read the stored values from the database inside the transaction. The event log's "before" had the same weakness and now reads the database too
"Barcode 91 already belongs to Bananas"	In the JS hooks `get()` on a JSON field returns raw bytes (91 = "[")	Parse `getString()` (already a known gotcha)
A Draft showed outdated reasons	Reasons were computed when the product was created, before its units existed	Recomputed whenever a unit changes
400 answer lost the list of problems	PocketBase rewrites the data of thrown errors	The endpoint rolls back, then answers the list itself
13 tasks after loading sample data	Each product passed through Draft, so the Drafts task was created and closed 11 times	Reopen the last Drafts task (DL-71)
Migrations could not be reverted	Hard-deleting units re-validated their Active product	Hard deletes (superuser only) skip that refresh
Reversibility test missed the P1 migrations	It counted only names starting `17912`	Counts every `1791…` migration
Backups test failed sometimes ("already running")	The real every-minute backup job started a nightly backup during the test (any time after 02:30)	Test hubs switch that job off (`CHEDAM_TEST_NO_BACKUP_CRON`) and call the tick themselves; 5/5 passes
Review: nested packs, inner pack resized	A case did not follow when its inner pack changed size; nested packs of "each" items allowed fractions; an inner pack could be removed	Fixed, 3 tests added

8. Session 3: P1 step 1 screens (commit `266962e`)

Screen / file	What
`client/src/screens/Products.svelte`	List with search by name, barcode or PLU, category and status filters, main unit's price, margin % (costs.view only), Draft reason
`client/src/screens/ProductEdit.svelte`	Product form: fields shown by type (weighed, perishable, regulated, imported), selling units with nested packs, barcodes, live cost / margin / markup / per-item price (costs.view only), "Save and make active" (prices.edit), archive, price history, Draft reasons. Read-only without catalogue.edit
`client/src/screens/Categories.svelte`	Add, edit, remove (only when empty), colour, order, till visibility, return window
`client/src/screens/Tax.svelte`	Classes and what they charge in the store's province, rates with dates (in use / upcoming / ended), add rate, set end, add custom class; "pending accountant review" notice
`client/src/lib/catalogue.js`	Money in cents, margins, device-made ids, nested pack quantities
Home, `session.svelte.js`, `App.svelte`	Products tile; the Drafts task opens the Drafts list; addresses `#products`, `#categories`, `#tax`
`tools/devhub/dev-hub.sh`	Local dev hub with the sample store on 127.0.0.1:8095 for checking screens in the browser

Checked in the browser (local dev hub)

- Paired the preview browser with a dev pairing code; signed in as Mira Manager (sample PIN).
- Drafts task → "Show" → Drafts list → mango form shows "'Single' needs a price of at least $0.01".
- Price $1.99 and a 4-pack at $6.99 added: the form showed "= 4 items", cost $3.80, margin 45.6%, markup 83.9%, $1.75 per item. "Save and make active" → Active.
- Tax screen: Standard = GST 5% + PST 7%, GST only, Zero-rated 0%, Exempt; both rates "in use".
- Categories: added "Spices".

- Signed in as Cal Cashier: no Add/Categories buttons, no margins in the list; product form read-only with no cost or margin; barcode 2000000000053 found Sparkling water.
- Phone width (375 px): no sideways scrolling. No console errors.

Failed cases and fixes (session 3)

Symptom	Cause	Fix
Price field showed "0.001.99" after typing	A unit with no price was filled with "0.00" as text	Empty prices and costs start blank
Typing `#tax` while a product was open did nothing	The open screen won over the address (also in P0 screens)	An address that differs from the open screen now wins
"1 products"	Plural not handled	Fixed
Zero-rated product said "No tax"	The class treatment was not used	Shows "0% (zero-rated)" or "No tax (exempt)"
Preview could not start `bash`	The preview runner needs the full path	Git's bash by its full path (Program Files, Git, bin, bash.exe)

9. Session 4: step 1 screens deployed; P1 step 2 (stock)

Deploy (Sreya: "proceed")

- `bash hub/scripts/deploy.sh --sample-data`: version `dev-42ea867`, no errors; the Pi serves the new app (checked the built file over HTTPS from the PC).

Hub side (commit `aa956dd`)

File	What
`hub/pb_migrations/1791300004_p1_stock_schema.js`	`stock_levels` (loose + sealed packs, on hand), `stock_lots` (expiry, cost per base unit), `stock_movements`, `stock_counts`, `stock_count_lines`; read-only through the generic API
`hub/pb_migrations/1791300005_p1_stock_reference.js`	Permissions `stock.receive`, `stock.adjust`, `stock.count` (staff, manager), `stock.approve` (manager); settings `stock.reasons`, `stock.approval_value_cents` = 2500
`hub/pb_hooks/lib/stock.js`, `lib/stock_http.js`, `stock.pb.js`	Receive, adjust/damage/loss, approve/reject, pack break/make, counts, product stock view, shrink report; one transaction each; `op_id` makes retries safe
`hub/pb_hooks/lib/tasks.js`	Shared rule-task helper (Drafts task, stock approvals)
`hub/pb_migrations_dev/1791300101_dev_sample_stock.js`	Sample deliveries, cola in cases, an expired milk lot
`hub/tests/step12-stock.test.mjs`	56 tests (the test hub can now leave out a sample file: `skipDev`)

Screens (commit `0eac2df`)

- **Stock** list (in stock / low / out), **product stock** (sealed and loose, lots by expiry, history, damage / loss / adjust with reason and photo, open or make packs), **Add stock** (camera, USB scanner or typing; "packs or singles?"; unknown barcode → new product form with the barcode; last cost with ▲/▼ %; lot and expiry; the list stays on the phone until saved), **Counts**, **Approvals**, **Shrink report** with CSV export.
- Camera: `client/src/lib/scan.js` + `components/Scanner.svelte`: BarcodeDetector, else ZXing (`@zxing/library` 0.23.0) loaded only when needed (DL-78).

Checked in the browser (local dev hub, signed in as Sam Staff, then Mira Manager)

- Stock list showed the sample levels (cola 82).
- Add stock: cola typed twice → quantity 2; water barcode → "Packs or singles?" → 6-pack at last cost $3.60; unknown barcode → new product form with the barcode filled in → saved as Draft → "Add it to the delivery" → back in the list; saved 3 lines; the list cleared.
- Camera button: the preview browser blocks cameras; the app showed "Camera permission was refused … or type the code". **Real phones still to test (R5).**
- Staff loss of 3 vape pods ($27) → "a manager approves it"; manager approved → stock 2; shrink report $27.00, Theft.
- Count: cola counted 3 sealed cases + 10 loose = 82, expected 84 (82 + 2 received), variance –2; submitted, approved; stock 82, lots add up to 82.
- Phone width: no sideways scrolling after a fix; the list survived a page reload.

ZXing check without a camera

A script drew EAN-13 barcodes into grayscale pictures and decoded them the way the app does: clean and moderately noisy pictures were read correctly; very heavy noise was not (expected).

Failed cases and fixes (session 4)

Symptom	Cause	Fix
Every stock request failed: "stockCtx is not defined"	Each PocketBase JS handler runs in its own VM; helpers at the top of a `.pb.js` file are not visible	Helpers moved to `lib/stock_http.js`
Breaking a 12-pack gave 12 *sealed singles*	The 12-pack "contains Singles"; the code put inner units back as sealed	A pack whose inner unit is a single or by-weight unit opens into loose stock
My reject test: "already decided"	That write-off ($18) was under the $25 limit and posted at once	The test receives more stock first and writes off $30
ZXing tried QR, Data Matrix, Aztec… on every frame	`decode()` resets the format list	1D reader used directly with the retail formats
`0628915120011` read as `628915120011`	A 0-led EAN-13 is also a valid UPC-A; readers may report either	Lookup and uniqueness treat both forms as one code (DL-77), 3 tests
Code box squeezed on a phone; "1 entries"	Layout and plural	Fixed

Tests: **464 passed, 0 failed** (12 suites).

10. Session 5: step 2 deployed; P1 step 3 (till and checkout)

Deploy (Sreya: "ok. proceed")

- `bash hub/scripts/deploy.sh --sample-data`: version `dev-aba2da4`, no errors. On the Pi: 11 stock levels (258.75 base units), 14 lots, integrity ok, PocketBase 52 MB.

Hub side (commit `0c278bf`)

File	What
`hub/pb_migrations/1791300006_p1_sales_schema.js`	`tills`, `cash_movements`, `sales`, `sale_lines`, `payments`, `tax_exemptions`, `holds`, `soft_holds` (read-only through the generic API)
`hub/pb_migrations/1791300007_p1_sales_reference.js`	Permissions `sales.sell`, `sales.discount`, `sales.tax_exempt` (cashier, manager), `sales.approve`, `sales.void`, `till.manage` (manager), `sales.view` (manager, accountant); settings: payment methods, cash rounding, US rate, discount and override limits, soft holds, exemption reasons, numbers, float, denominations, variance task

`hub/pb_hooks/lib/pricing_core.js`	Pure sale arithmetic: cart discount spread before tax, tax per type rounded once, tax included, cash rounding
`hub/pb_hooks/lib/sales.js`, `lib/tills.js`, `lib/sales_http.js`, `sales.pb.js`	Quote, complete, approvals, void, holds, soft holds, exempt report; tills: open, cash, close, Z report
`hub/tests/step13a-pricing.test.mjs`, `step13-sales.test.mjs`	16 + 75 tests

Decisions DL-79..85 in `docs/P1-mini-spec.md`.

Screens (commit `456a609`)

Sell (grid, search, scanning, weight, packs or singles, ID check, line edit, cart discount, tax exempt, hold/recall, training, live quote, manager PIN, payments), **Till** (float, drops, pay-outs, no-sale, close, Z report, till settings), **Sales** (find, receipt, reprint, void, exempt report), **Receipt**.

Checked in the browser (local dev hub, Cal Cashier)

- Opened till 1 with $20×5, $5×4, $1×10 = $130.00.
- Basket: cola scanned twice, chips, PLU 4011 bananas 1.235 kg → **$9.88, cash $9.90** (GST 35 ¢, PST 21 ¢, deposits 20 ¢), as worked by hand. Paid $20 → change **$10.10**, receipt S-000001.
- Vape pods: "Check ID … 19+" → ID checked; 20% cart discount → $17.91 → "Manager approval needed: Cart discount above 10%" → Mira Manager's PIN → card ****4242 approved → S-000002, "You saved $4.00".
- Close: expected $139.90, counted $139.00 → Z report "Short $0.90"; GST $1.15 on $22.96, PST $1.33 on $18.97, cash $9.90, card $17.91.
- Phone width: no sideways scrolling.

Failed cases and fixes (session 5)

Symptom	Cause	Fix
Approved sales still refused (403)	PocketBase's shared store keeps a copy of a JS object; adding an approval to the map afterwards was lost	Write the map back after every change
Soft-hold test: nothing held	My setup sale was on the manager's device, which had no open till	Test sells on the cashier's till
Training sales missing from the Z report	They were not linked to the till	Linked when a till is open, still kept out of money and stock
My rounding test failed	My arithmetic: 18.549 rounds to 19	Test uses 18.49 → 18
A shell edit failed with "unexpected EOF"	Quotes inside a here-document	Wrote those files with the file tool; nothing half-written
Cart column pushed off screen	The scrolling category row widened its grid column	`min-w-0` on the column
A typed banana weight went into the scan box	`autofocus` does not work on prompts that appear later	Prompts take the focus themselves
Approval dialog build warning	`bind:this` on a plain variable	`$state(null)`

Tests: **555 passed, 0 failed** (14 suites).

Failed cases and fixes

Symptom	Cause	Fix
A device worked without the certificate	The person tapped through Chrome's warning	Amber warning bar in the app (DL-62)
"Hub gone" with no reboot	Laptop SSH tunnel dropped for a few seconds	Correctly ignored (boot id unchanged)
Writer crashed after pull 1	4 s timeout too short for a freshly booted Pi Zero listing ~17,000 records	60 s, 3 attempts; confirmed writes kept in a file
My journal query found nothing	The script logs in UTC, the Pi in Pacific time	Queried the right local time

Hub came back without Wi-Fi	Netplan Wi-Fi files emptied by a power cut during boot	`chedam-netguard` (DL-60)
`\n` turned into a real line break in a Python edit	Quoting layers	Fixed the line directly
23 confirmed writes lost	SQLite `synchronous=NORMAL`	`synchronous=FULL` (DL-61)
Intermittent test failure "one urgent task"	Update/backup result applied in two writes, by two callers at once	One transaction per result (`85da79a`)

Next

1. Sreya: go-ahead to deploy step 3 (selling) to the Pi; then try a sale on the iPhone/Android and the PC.
2. Camera test on real phones (R5, DL-63): Add stock → Scan with camera, and Sell → 📷.
3. P1 step 4: offline selling (tills keep selling when the hub is unreachable).
4. Open: Q1 accountant review (tax classes, deposits, exemption reasons; blocks the P1 release), Q2 return defaults; GST/PST numbers to enter in the business profile (they print on receipts).

2026-10-05 (Monday, October 5, 2026)

P0 steps 1-9 built and deployed; gate: 3-device login, update + rollback, backup + restore, memory under load passed: hub base, schema + event log, access, devices, forgot PIN, client app; work log started

Summary

- Full health check of the re-flashed Pi: healthy, OS up to date, no hub software yet.
- Restored admin access (the sudoers rule is lost on every re-flash).
- Changed `setup-hub.sh` so the display setting can be switched **both ways** (monitor on or headless) and re-runs leave the boot config unchanged.
- Ran hub setup with `--keep-display`, rebooted and verified everything. HTTPS works by name and by IP.
- iPhone 13 Pro and Android 9 installed the new "Chedam Store CA" and opened the hub page without warnings ("Chedam hub is running").
- Started this cumulative work log (`Work Log/`).
- P0 step 1 (hub base) is complete.
- P0 step 3 (access) is complete: permission engine with role templates, overrides and BR-33, PIN sign-in with lockout, owner recovery code, "Who can do this?". 92 more tests (124 in total) passing; deployed and checked on the Pi.
- P0 step 4 (devices) is built and tested on the PC (67 new tests, 192 in total): pairing by code/QR or by approval request, device key checked on every request, name-then-PIN sign-in on paired devices only, one person per device, device manager (approve, lock, sign out, rename, assign, remove), certificate guide. Deployed to the Pi and checked; pairing the real phones is next.
- Forgot PIN decided and built: a manager sets a temporary PIN, and the person chooses their own at the next sign-in.
- P0 step 5 (client shell) is built and tested on the PC: an installable Svelte app that opens offline, with the connectivity bar, name-then-PIN sign-in, choosing a new PIN, owner sign-in, idle auto-lock, large text and high contrast, and the device manager with QR codes. It replaces the temporary pages. 213 hub tests pass.
- P0 step 6 (setup wizard) is built, tested and deployed: one-time setup code on a new hub, owner and first device in one step, recovery code, business profile, logo and receipt, team, storage, features, references, backups; skipped steps become tasks. 251 hub tests pass.
- P0 step 7 (backups) is built and tested **on the Pi** with a disk image: encrypted, verified backups, restore test, and a real restore in 2-3 s that keeps the HTTPS certificate authority. Sreya's hard drive showed read errors, so nothing was written to it. 281 hub tests pass.
- **3 devices signed in as the owner** (iPhone, Android 9 phone, Surface laptop) after fixing a sign-in mix-up and the laptop certificate. Issue logged: the Android 9 phone runs Chrome 98, too old for the app's styles (Tailwind 4 needs Chrome 111+), so its screens are misaligned.
- P0 step 8 (hub health page) is built and deployed; 302 hub tests pass.
- P0 step 9 (updates) is built and tested on the Pi: a signed install in 1.7 s, and a broken update rolled back automatically in 4 s; tampered packages are refused. 323 hub tests pass.
- First backup to the new SD card made by Sreya and verified independently; **0.9.0 published** to `chedam-releases` and downloaded and verified by the Pi from GitHub.
- P0 gate, memory under load: 10 devices for 5 minutes peak at **108 MB** (budget 150 MB). A stress burst first reached 194 MB; per-device rate limits now hold it at 120 MB.
- Re-tested the step 3 bugs and issues: all fixed. A third copy of one bug (recovery code) was found and fixed.
- P0 step 2 (schema v1 + event log) is complete: 14 tables, reference data, dev sample store, a transactional append-only event log, 32 automated tests passing, deployed to the Pi with a backup first.

Naming note: the local folder `chedam-updates/` in the project is Sreya's YouTube and notes repo (gitignored). It is **not** the update channel. Nothing is renamed; the public update-package repo name will be decided when step 9 (Updates) starts.

1. Health check (before any change)

One SSH session ran these checks:

```
whoami; hostname; hostname -I; cat /proc/device-tree/model; grep PRETTY /etc/os-release; uname -r
uptime; sudo -n true; free -m; df -h / /boot/firmware; lsblk
vcgencmd measure_temp; vcgencmd get_throttled; vcgencmd measure_volts core
timedatectl; ip route; ping -c2 1.1.1.1
apt list --upgradable; dpkg-query -W <packages>; systemctl is-active ...; systemctl --failed
cat /boot/firmware/config.txt /boot/firmware/cmdline.txt
```

Area	Result
Hardware	Pi Zero 2 W Rev 1.0, 33.2 °C, throttled 0x0, core 1.2875 V
OS	Debian 13 trixie, kernel 6.18.50+rpt-rpi-v8, 0 pending updates
Uptime	1 day 12 h, load 0.00, no failed units
Memory / disk	415 MB total, 273 MB available / 119 GB card, 2.3 GB used
Network	192.168.50.101 via 192.168.50.1, internet ping 7 ms
Time	NTP synced, America/Vancouver, no RTC
sudo	**Needed a password** (re-flash reset it)
Hub software	Not installed: no caddy, chrony, ufw, sqlite3 or `/opt/chedam`
Display	`video=HDMI-A-1:1280x720@60D` present (monitor in use)

2. Admin access

Failed case: `Permission denied (publickey)` when restoring sudo

- The `ssh -t chedam "…sudo tee…"` command was typed **inside an SSH session on the Pi** (prompt `chedam@chedam:~ $`).
- So the Pi tried to SSH into itself and had no key for that. Re-flashing was not needed.
- Fix: run only the inner command on the Pi:

```
echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmod 440 /etc/sudo
sudo -n true && echo OK
```

3. Display becomes a switch (DL-24 updated)

Sreya's requirement: the monitor is optional on the Zero 2 W pilot, but the option must stay. Pi 4 production hubs will use a display to operate the hub directly.

Change in `hub/scripts/setup-hub.sh` (boot config section):

- Removes any earlier chedam block, then writes a block between `# chedam-begin` and `# chedam-end`.
- `--keep-display`: display driver `vc4-kms-v3d` and display detection **on**, Bluetooth off.
- No flag: headless. The display driver and detection are off, `gpu_mem=16`, Bluetooth off; this saves ~50 MB of RAM.
- Audio and camera detection are always off.

```
sudo bash ~/hub/scripts/setup-hub.sh --keep-display   # monitor on (current)
sudo bash ~/hub/scripts/setup-hub.sh                  # headless
sudo reboot                                           # after switching
```

Failed case: an extra `[all]` line on every run

- The test (switching off, on, off, on on a copy of `config.txt`) worked, but each run appended another `[all]` line. It was harmless but untidy.
- Fix: moved `[all]` inside the marked block. The original config (`config.txt.chedam-orig`, saved by the first run) was restored and setup run twice. Both runs gave the same checksum (`d91ff929`), and the only differences from the original were audio off, camera detection off and the chedam block.

4. Hub setup on 192.168.50.101

```
scp -r scripts system chedam:hub/
# on the Pi: download + verify PocketBase
wget pocketbase_0.40.4_linux_arm64.zip checksums.txt
grep linux_arm64 checksums.txt | sha256sum -c -      # OK
sudo apt-get update
sudo bash ~/hub/scripts/setup-hub.sh --keep-display
sudo systemctl reboot
```

Verification after reboot:

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon, ufw active; no failed units
Memory	PocketBase 28 MB + Caddy 37 MB = 65 MB (budget 150 MB, NFR-08); 264 MB available with display on
Watchdog	`RuntimeWatchdogUSec=14s`
Time	chrony, 0.0000007 s from NTP, leap normal
Temperature	38.6 °C, throttled 0x0
Display	`/dev/dri/card0` present
Firewall	22, 80, 443/tcp and 5353/udp allowed (IPv4 + IPv6)
CA download	`http://chedam.local/ca.crt` → 200, "Chedam Store CA - 2026 ECC Root", valid to 2036-08-13
HTTPS	`https://chedam.local/api/health` and `https://192.168.50.101/api/health` → 200
Certificate check	`openssl s_client … -CAfile ca.crt` → Verify return code 0, with and without SNI
HTTP	Redirects to HTTPS (302)
Caddy errors since boot	0

PC test commands (Git Bash; `--resolve` avoids Windows mDNS problems, and `--ssl-no-revoke` avoids the Schannel revocation error):

```
curl --resolve chedam.local:80:192.168.50.101 -o ca.crt http://chedam.local/ca.crt
curl --ssl-no-revoke --resolve chedam.local:443:192.168.50.101 --cacert ca.crt https://chedam.local/a
curl --ssl-no-revoke --cacert ca.crt https://192.168.50.101/api/health
```

5. Phones (risk R5, first part)

Device	How	Result
iPhone 13 Pro, iOS 26.6.1	Safari `http://chedam.local/ca.crt` → install profile → Settings > General > About > Certificate Trust Settings → on	`https://chedam.local` shows the placeholder, no warning
Android 9	Chrome `http://192.168.50.101/ca.crt` → install for VPN and apps	`https://192.168.50.101` shows the placeholder, no warning

The camera test over HTTPS comes later, when the client app exists.

6. Work log started

- Folder `Work Log/`: one Markdown source per day in `Work Log/entries/YYYY-MM-DD.md`.
- `tools/worklog/build_worklog.py` builds `Work Log/Chedam-Work-Log-YYYY-MM-DD.pdf` with **all** days, newest first.
- Earlier PDFs are kept, so each day has its own complete snapshot.

```
python tools/worklog/build_worklog.py             # newest day
python tools/worklog/build_worklog.py 2026-10-05 # a specific day
```

The days 2026-10-02 and 2026-10-03 were written up from the session records and `docs/P0-pi-setup-log-2026-10-03.pdf`.

7. P0 step 2: schema v1 and event log

Spec basis: Section 10 (data model, event log), Section 13 (build process: migrations and sample data first), NFR-20 (reversible migrations), BR-30 (UTC), BR-33 (managers cannot grant more than they hold).

Files added

File	Purpose
`hub/pb_migrations/1791200001_p0_schema_v1.js`	13 new tables + the PocketBase `users` collection extended: business, locations, settings, modules, permissions, roles, permission_overrides, devices, storage_areas, tasks, events, backups, updates. Common fields on every table. API rules all closed (superuser only) until step 3
`hub/pb_migrations/1791200002_p0_reference_data.js`	22 modules (3 core always on, 15 switchable, 4 later), shop-type presets, 20 P0 permissions (5 owner-only), 5 role templates, 12 default settings
`hub/pb_migrations_dev/1791200100_dev_sample_store.js`	Dev-only sample store: "Demo Grocery", main location, grocery preset modules, 4 storage areas, 5 people (one per role, random unknown passwords), 1 setup task
`hub/pb_hooks/event_log.pb.js` + `lib/event_log.js`	Event log: one `events` row per create/update/delete, in the same transaction; actor and device stamped from the request; append-only
`hub/tests/step2-schema-eventlog.test.mjs`	32 automated tests against a throwaway local PocketBase
`hub/scripts/deploy.sh`	PC → hub deploy: copy, SQLite online backup + integrity check, rsync, restart, health, migrations, errors
`docs/conventions.md`	Data, event-log, migration and workflow rules for all phases

Design choices

- **Common fields:** `id` (a device may create it offline), `created_at`, `updated_at`, `created_by`, `updated_by`, `device_id`, `deleted_at`. `updated_by` was added beyond the spec's list, so each change records who made it.
- **The event log runs inside the same transaction** as the change (`onRecord*Execute` + `runInTransaction`). If the event can't be written, the change is rolled back, so the log never misses a write.
- **Actor and device are stamped by the hub** in request hooks. A client cannot forge `created_by`, and the device comes from the `X-Chedam-Device` header (verified in step 4).
- **Hidden fields** (`pin_hash`, `recovery_code_hash`, `key_hash`, `pairing_code_hash`) never reach the API or the log.
- **Role levels:** owner 100, manager 80, accountant 50, cashier 40, staff 30. Owner-only permissions: modules.manage, setup.run, owner.manage, backups.restore, updates.install.
- **Sample data is separate** from real migrations, so a store hub never receives demo records.

Commands

```
# local tests (PC, Git Bash)
PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/step2-schema-eventlog.test.mjs
# deploy to the Pi with sample data
bash hub/scripts/deploy.sh --sample-data
# verify on the Pi
ssh chedam 'sudo -u chedam-hub sqlite3 /opt/chedam/pb_data/data.db "select table_name, action, count(
```

Failed cases and fixes

Symptom	Cause	Fix
`pocketbase.exe: command not found` in Git Bash	That shell started before `C:\Tools` was added to PATH	Full path `C:\Tools\pocketbase.exe` / `PB_BIN`
`migrate down` hung	It waits for a y/N confirmation	Pipe `y` (`input: "y\n"` in the tests)
Python check query error (`ESCAPE expression must be a single character`)	Backslash escaping in an inline script	Filter with `substr(name,1,1) != '_'`
Sample migration: `Cannot read property 'forEach' of undefined`	A JSON field read with `get()` returns raw bytes	`JSON.parse(record.getString("value"))`
Test stop step hung	Waited for the exit of a process that had already exited	Check `exitCode`/`signalCode` first
Create returned 400 "Something went wrong"	`e.request` doesn't exist in PocketBase 0.40 JS hooks (found with `serve --dev` logs)	`e.requestInfo().headers["x_chedam_device"]` (names lowercased, `-` → `_`)
A Python edit put real newlines inside JS strings	Escaping in a heredoc	Fixed with a direct edit

Test results (PC, 32/32 passed)

- Schema/reference: 22 modules, 3 core on, 15 switchable, 20 permissions, 5 roles, owner holds all, manager holds no owner-only permission, 12 settings.
- Sample: business + location, 4 storage areas, 5 people, 1 task, 7 grocery modules on.
- Event log: device-made id accepted; forged `created_by` overwritten; device stamped; create/update/soft-delete/delete each logged with correct before/after/changed; delete keeps the last state; no PIN hash, password or tokenKey in the log; events can't be edited or deleted, even by a superuser.
- Atomicity: with the event write forced to fail, the create fails and the record is not in the table.
- Migrations: down 3 and up again both clean.

On the Pi (after deploy)

Check	Result
Pre-deploy backup	`pre-deploy-<UTC>.db`, integrity ok
Migrations applied	schema v1, reference data, dev sample store
Rows	modules 22, permissions 20, roles 5, settings 12, business 1, locations 1, storage_areas 4, users 5, tasks 1
Events	78 = every write above (incl. 7 module updates for the grocery preset)
Secrets in events	0
SQLite journal	WAL
PocketBase memory	42 MB (budget 150 MB)
Errors in the journal	none

8. P0 step 3: Access

Spec basis: FR-1.03 (owner password, PIN, printed recovery code), FR-1.04 (people with role and PIN), FR-1.09 (owner signs in on any device), FR-1.10 (role templates, per-person overrides with end date, "Who can do this?"), NFR-11 (hub-side permission checks, PIN lockout after 5 tries, auto-lock), BR-33 (managers can't grant more than they hold).

Files added or changed

File	Purpose

`hub/pb_migrations/1791200003_p0_access.js`	PIN and recovery code become bcrypt password fields (hidden); `pin_set` and `recovery_code_created_at` added; API rules set on all 14 tables; token lifetime 12 h. Reversible
`hub/pb_hooks/lib/access.js`	Permission engine: effective permissions (role + allow overrides − deny overrides; owner = all), the per-table access map, protected fields, self/assignee field limits, BR-33 checks for people, roles, overrides, modules, settings, business
`hub/pb_hooks/lib/auth.js`	PIN rules, lockout counter, recovery code generator, secret check
`hub/pb_hooks/access.pb.js`	Guard on every list/view/create/update/delete; password sign-in lockout; token refresh check; endpoints (below)
`hub/pb_migrations_dev/1791200101_dev_sample_pins.js`	Dev-only PINs for the 5 sample people (test values; dev hubs only)
`hub/pb_hooks/lib/event_log.js`	Fix: put the original app back after the transaction (see failed cases)
`hub/tests/lib/hub.mjs`, `step3-access.test.mjs`, `run-all.mjs`	Shared test harness, 92 access tests, one-command runner
`docs/conventions.md`, `docs/P0-mini-spec.md`	Access rules for later phases; decisions DL-28 to DL-32

Endpoints

Method and path	Who	What
`GET /api/chedam/auth/pin-users`	anyone on the LAN (step 4: paired devices only)	Names for the till sign-in screen
`POST /api/chedam/auth/pin`	anyone	`{user, pin}` → sign-in token; wrong tries count toward lockout
`POST /api/chedam/users/{id}/pin`	yourself, or users.manage above them	Set or change a PIN (4-6 digits, no repeats or straight runs)
`POST /api/chedam/users/{id}/unlock`	users.manage above them	Clear a lockout
`POST /api/chedam/owner/recovery-code`	owner	New printable code, shown once (format `XXXXX-XXXXX-XXXXX-XXXXX`)
`POST /api/chedam/auth/recover`	anyone with the code	`{email, code, new_password}` → new password + sign-in; the code is used up
`GET /api/chedam/access/me`	signed in	My role and effective permissions
`GET /api/chedam/access/users/{id}`	self or users.view	Someone's effective permissions and where each comes from
`GET /api/chedam/access/who-can/{code}`	users.view	"Who can do this?" (via role or override, with end date)

Who may do what (summary of the access map)

- **Anyone signed in:** reads the business profile, locations, settings, modules, permissions, roles and storage areas.
- **People:** list with users.view; a person can view and edit their own record (name, email, phone, language, accessibility, password only); add/edit others needs users.manage; giving or changing roles needs access.manage; no hard delete.
- **Overrides:** access.manage only; never on yourself; only on people below you; never grant what you don't hold; owner-only permissions only by the owner.
- **Roles:** templates can't be deleted; only the owner edits templates; the owner role always holds everything; managers create and edit only roles below their own level.
- **Modules:** owner only; only `enabled` can change; core can't be switched off; "later" modules can't be switched on; who/when stamped.

- **Settings:** `security.*` owner only; keys can't be renamed.
- **Audit log:** events.view. Tasks: tasks.view / tasks.manage (the assignee may update status/note). Devices: devices.view / devices.manage. Backups and updates: read-only through the API.

Failed cases and fixes

Symptom	Cause	Fix
No way to check a bcrypt PIN in JS hooks	`$security` has no bcrypt	Use a PocketBase `password` field: `record.getRaw("pin").validate(plain)`
Owner recovery sign-in failed: `failed to delete all previous MFAs … transaction has already been committed`	The event-log wrapper replaced `e.app` with the transaction and never restored it, so PocketBase's clean-up after a password change used a finished transaction. **This would have broken every password change.**	Restore `e.app` in a `finally` block after `runInTransaction`
5th wrong password returned 400, not 423	The failure was counted, but that same response didn't report the new lock	After counting, re-read the person and answer "locked" (same as PIN sign-in)
Assumed PocketBase leaves `e.record` empty on a wrong password	Traced with debug lines: the hook gets the person and `e.next()` checks the password	Removed the extra lookup and the debug lines
Step 2 test still sent `pin_hash` and reversed only 3 migrations	The schema changed in step 3	Test uses the `pin` field, checks for no `$2a$` hashes, and reverses every Chedam migration it finds
`sed` edits to the test harness didn't apply	Escaping in shell quoting	Direct edits

Test results (PC)

`node hub/tests/run-all.mjs` → step2 **32/32**, step3 **92/92**.

Step 3 covers: - PIN sign-in, the PIN list showing no secrets, and wrong PINs refused. - Lockout after 5 wrong PINs and after 5 wrong passwords (423), and unlock by a manager but not a cashier. - The PIN rules. - Who can read and change each table. - Managers adding cashiers but not owners or other managers, and editing people below them but not the owner. - Role changes, and the rule that you can't change your own role. - Suspending someone signs them out at once. - Overrides: allow, deny, past and future end dates, owner-only, and "can't grant what you don't hold". - Role templates, the owner role, and deleting custom roles. - The module, setting and business rules. - The recovery code: shown once, case-insensitive, used once. - Who-can and my-permissions. - The audit trail: lockouts, overrides and module switches are logged, and no hashes appear anywhere.

On the Pi (after deploy, checked on the Pi itself)

Check	Result
Deploy	Backup + integrity ok; migrations `1791200003_p0_access.js` and `1791200101_dev_sample_pins.js` applied; no errors
PIN list	5 people
Signed-out request	401
PIN sign-in time	0.50 s (bcrypt cost 10 on the Pi Zero 2 W)
`access/me` for the cashier	role cashier, permissions `tasks.view`
Cashier lists people	403
bcrypt hashes in the event log	0
PocketBase memory	44 MB

The dev PINs are in the dev seed file only and are not repeated in this log.

9. Re-test of the step 3 bugs and issues

Sreya asked which 2 bugs and 2 issues came up in step 3, whether they were fixed, and to test them again.

#	Kind	What	Status after re-test
1	Bug	Event-log wrapper left `e.app` pointing at a finished transaction, so the owner's recovery sign-in (and every password change) failed	Fixed. Tests "recovery code + new password signs the owner in" and "new password works" pass
2	Bug	The 5th wrong password answered 400 instead of 423 (locked)	Fixed. Test "5 wrong passwords lock the owner too" passes
3	Issue	PocketBase JS hooks have no bcrypt function	Solved with PocketBase `password` fields; PIN tests pass
4	Issue	The step 2 test was out of date after the step 3 schema change	Updated; step 2 suite passes (now 33 tests)
5	**Bug found now**	Same as bug 2 on the **recovery code** path: the 5th wrong code answered 400, not 423	Fixed: one shared `auth.failAndThrow()` now serves PIN, password and recovery code. New test "5th wrong recovery code answers 423"

Baseline before any change: `node hub/tests/run-all.mjs` → 32/32 and 92/92.

10. P0 step 4: Devices (FR-1.07, 1.08)

Files added or changed

File	Purpose
`hub/pb_migrations/1791200004_p0_devices.js`	`paired_via`, `paired_at` on devices; generic create closed; settings `devices.pairing_code_minutes` (10), `devices.online_seconds` (120), `devices.max_pending_requests` (10). Reversible
`hub/pb_hooks/lib/devices.js`	Device check for every `/api/` request, pairing codes, keys, last seen, device view
`hub/pb_hooks/devices.pb.js`	Middleware + endpoints (below)
`hub/pb_hooks/access.pb.js`	Password and PIN sign-in need a paired device (owner excepted); sign-in records who is on the device; assigned devices; recovery-code fix
`hub/pb_hooks/lib/access.js`	Devices: generic create closed, protected fields, BR-33 on assigning
`hub/pb_hooks/lib/auth.js`, `lib/event_log.js`	`failAndThrow()`; device stamped only when verified
`hub/pb_public/`	Temporary pages until the Svelte client (step 5): `device-setup.html` (certificate guide, works over HTTP), `pair.html` (pair, then pick your name and enter PIN), `devices.html` (device manager with QR code), `assets/` (shared CSS/JS, vendored QR library, `VENDORED.md`)
`hub/system/Caddyfile`	HTTP serves the guide and its CSS; `/hub.json` gives the hub IP for QR codes
`hub/scripts/deploy.sh`	Also updates the Caddy config when it changed (keeps the IP, validates, keeps a copy, reloads)
`hub/tests/lib/hub.mjs`, `step2…`, `step3…`, `step4-devices.test.mjs`	`pair()` helper and automatic device headers; existing suites now sign in on paired devices; 67 new tests
`docs/P0-mini-spec.md`, `docs/conventions.md`	DL-33 to DL-38, device rules, the open "forgot PIN" item

Endpoints

Method and path	Who	What
`POST /api/chedam/devices/pairing-code`	devices.manage	`{name, type}` → one-time code `XXXX-XXXX` (10 min)
`POST /api/chedam/devices/pair`	new device	`{code}` → device id + key, approved
`POST /api/chedam/devices/request`	new device	`{name, type}` → id + key, waits for approval
`GET /api/chedam/devices/me`	the device	Its status and who is signed in (works while pending or locked)

`POST /api/chedam/devices/me/sign-out`	the device	Sign out whoever is on it
`GET /api/chedam/devices`	devices.view	Device manager list: online, signed in, assigned, version, last seen, browser
`POST /api/chedam/devices/{id}/approve, lock, unlock, sign-out, revoke`	devices.manage	Device manager actions
`PATCH /api/collections/devices/records/{id}`	devices.manage	Rename, type, assign person or printer

Sign-in flow (Sreya: "select name and then PIN")

The device is paired once. After that the sign-in page shows the people's names as large buttons. Tap your name, enter your PIN on the keypad and press OK. A wrong PIN shakes the keypad and shows "Wrong name or PIN". After 5 wrong tries the person is locked (423). "Not you? Back to names" returns to the list. The owner also has "sign in with email and password", which works on any device. "Forgot your PIN?" shows the current options (see the open item in the mini-spec).

Commands

```
PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/run-all.mjs
npm pack qrcode-generator@2.0.4      # vendored QR library; integrity checked against the npm registr
```

Failed cases and fixes

Symptom	Cause	Fix
Shell edit script stopped with "unexpected EOF while looking for matching quote"	Quotes and backticks of JS code inside a bash heredoc	Wrote the Python edit script to a file and ran it
Helpers at the top of `devices.pb.js` would be undefined in handlers	Each PocketBase JS handler runs in its own VM and cannot see top-level variables of the `.pb.js` file	Moved them into `lib/devices.js` (loaded with `require` inside each handler)
Test "key and code hashes never in API output" failed	The test read the record as **superuser**, and PocketBase shows hidden fields to superusers on purpose	The test now reads as the manager (who never sees them); superuser = development admin only
"Paired as …" message vanished at once	The status refresh after pairing cleared the message	Show the message after the refresh
PIN typed by the browser test tool did nothing	The tool inserts text without key events (a real keyboard sends them)	Tested by tapping the on-screen keypad, as on a phone
Deploy to the Pi blocked	Deploying to the hub needs Sreya's approval in this session	Sreya approved; deployed

Test results (PC)

`node hub/tests/run-all.mjs` → step2 **33**/**33**, step3 **92**/**92**, step4 **67**/**67** (192 in total).

Step 4 covers: - **Pairing codes:** who may make them, the format, lower case with a space accepted, single use, expiry, approved by the code's maker. - **Device keys:** unknown, wrong or missing keys refused (401); keys never in API output or the log. - **Ask to join:** pending until approved; the cap on open requests (429). - **Device needed:** no PIN list or PIN sign-in without a device; a cashier's password sign-in or token without a device refused; the owner works on an unpaired browser. - **One person per device:** the earlier token is refused, tokens can't move devices, sign-out. - **Device manager:** list (online, signed in, version, own device marked); sign-out, lock (423)/unlock, rename; protected fields; no generic create. - **Assigned devices:** list and sign-in limited; BR-33 on assign and remove. - **Revoke:** device and token refused, no double remove. - **Recovery:** the 423 fix, and owner recovery on a till. - **Log:** last-seen writes throttled (≤1 event for 10 requests), lock/unlock/revoke logged with actor and the manager's device, no hashes. - **Guessing:** 20 wrong codes pause pairing (429).

Browser check (local dev hub, PC)

Paired with a code from the URL. The name list showed 5 sample people. A wrong PIN gave "Wrong name or PIN" and the right PIN signed in (manager), with the Device manager link shown. The device manager showed the till online with

its version and browser. Pair a new device showed the QR code, the code and a countdown. At phone width (375 px) the certificate guide opened the Android steps automatically, with no sideways scrolling.

On the Pi (after deploy, checked from the PC over the LAN)

```
bash hub/scripts/deploy.sh --sample-data
```

Check	Result
Deploy	Backup integrity ok; `1791200004_p0_devices.js` applied; Caddy config updated for 192.168.50.101 and reloaded (old copy in `backups/`); no errors
Certificate guide over plain HTTP	`http://192.168.50.101/device-setup.html` 200, its CSS 200; other HTTP pages still redirect to HTTPS (302)
`https://192.168.50.101/hub.json`	`{"ip":"192.168.50.101","name":"chedam.local"}`
Pages over HTTPS	`pair.html`, `devices.html`, `assets/qrcode.js` 200
PIN list without a device	403
Made-up device header	401
Wrong pairing code	400
Device settings	3 `devices.*` settings present
PocketBase memory	40 MB (budget 150 MB)
Temperature	37.0 °C

11. Forgot PIN: temporary PIN (Sreya's decision)

Sreya: "manager reset temporary and user need to reset again".

- A PIN set by someone else (manager with users.manage above the person, or superuser) is **temporary**. It works for 24 h (`security.temp_pin_hours`).
- With a temporary PIN, the hub allows only these until a new PIN is chosen: setting your own PIN, `access/me`, `devices/me` and sign-out. Everything else answers 403 "Choose your own new PIN first." (`auth.pinChangeGate`, run in the request middleware).
- The new PIN must differ from the temporary one, and the PIN rules apply. An expired temporary PIN is refused with a clear message.
- `pin_must_change` and `pin_temp_expires_at` are protected fields, and both changes are in the audit log.
- Migration `1791200005_p0_temp_pin.js` (reversible). Tests: `step3b-temp-pin.test.mjs`, 14/14.
- In the app: the manager uses "Someone forgot their PIN" on the home screen; the person then sees "Choose your own PIN" (enter twice).

12. P0 step 5: client shell (FR-1.09, FR-12.01, NFR-16/17)

Hub side

File	Purpose
`hub/pb_migrations/1791200006_p0_connectivity.js`	Settings `network.check_url` (Cloudflare's 204 check) and `network.hotspot_ssids`
`hub/pb_hooks/lib/net.js`, `status.pb.js`	Check once a minute (cron); `GET /api/chedam/status` → `{internet: offline/router/hotspot, hub_time}`; open to waiting/locked devices too
`hub/pb_hooks/access.pb.js`	`access/me` also returns `pin_must_change`, large text, high contrast, language
`hub/tests/step5-status.test.mjs`	7 tests: open, values, hub clock, offline/online via the check address, guests can't force a check, waiting device

Client (`client/`)

Part	What
Stack	Svelte 5.57.1, Tailwind 4.3.3, Vite 8.3.2, qrcode-generator 2.0.4 (same integrity hash as checked in step 4). Exact versions; `npm audit` 0 vulnerabilities
Build	`npm run build` → `hub/pb_public/` (about 35 KB gzipped JS + 4 KB CSS). `deploy.sh` builds it before copying
PWA	Manifest, icons (drawn by `client/scripts/make-icons.mjs`, no image library needed), service worker generated with the exact file list
Screens	Pair (code / ask to join), Waiting (approval or locked), Names, PIN pad, Choose your own PIN, Owner sign-in, Home (modules switched on, "Someone forgot their PIN", My settings, Install), Devices (manager)
Connectivity bar	Online (router) / Online (hotspot) / Offline, "Hub not reachable", device clock warning (2+ min)
Auto-lock	After `security.auto_lock_minutes` idle: signed out on that device, name list returns
Removed	The temporary step 4 pages (`pair.html`, `devices.html`, `assets/chedam.js`, vendored `qrcode.js`). The certificate guide moved to `client/public/` and now opens the app with `/?pair=CODE`
Caddy	HTTP serves `/device-setup.html` and `/setup.css`

Browser check (local dev hub on the PC)

Check	Result
Session kept	The earlier paired till and Mira's session came back after the app update
Connectivity bar	"Online (router)"
Forgot PIN, end to end	Mira set a temporary PIN for Sam → Sam signed in with it → forced to "Choose your own PIN". A mismatch was caught, reusing the temporary PIN was refused, the new PIN was set and Sam reached his home screen (no Devices tile: staff)
Offline shell	Service worker active, 8 files cached. With the hub **stopped**, a reload still opened the app with the red "Hub not reachable" bar and a clear message
Recovery	Hub started again → bar back to "Online (router)"; Sam's session came back
Device manager	Mira → Devices → Pair a new device → QR code + code + countdown
Phone size, dark mode	Readable, no sideways scrolling
Certificate guide	Page and `setup.css` 200; continues to `/?pair=CODE`

Failed cases and fixes

Symptom	Cause	Fix
Step 2 test "12 default settings" failed (13)	It counted settings; step 4/5 add their own	Checks the 12 step-2 keys by name
Svelte warning: `pad` not declared with `$state`	Component reference in a plain variable	`let pad = $state()`
Offline reload showed a blank page	The module script is requested with an `Origin` header and the hub answers `Vary: Origin`, so the cached copy (stored without `Origin`) did not match	Cache look-ups use `ignoreVary: true` (our files are content-hashed)
Crash "Cannot read properties of null (reading 'user')" when signing out	Home still drew `s.me.user` for a moment after sign-out cleared it	Home renders only while someone is signed in
After the hub came back, the name list kept its error until the next 30 s poll	Nothing reacted to the hub returning	The app refreshes as soon as the hub answers again; the name list retries
First page load after restarting the dev hub was refused by the browser pane	The page was opened before the server was listening	Waited 2 s and opened it again

Test results (PC)

`node hub/tests/run-all.mjs` → step2 33/33, step3 92/92, step3b 14/14, step4 67/67, step5 7/7 (**213**).

On the Pi (after deploy, Sreya's go-ahead)

Check	Result
Deploy	App built (35 KB gzipped JS); backup integrity ok; migrations `…05_p0_temp_pin`, `…06_p0_connectivity` applied; Caddy updated and reloaded; no errors
App over HTTPS (IP and chedam.local)	`/`, `sw.js`, `manifest.webmanifest`, icons: 200
Certificate guide over HTTP	`device-setup.html`, `setup.css`: 200
`/api/chedam/status`	`internet: router`; hub clock matches
Wi-Fi name readable by the hub (`nmcli`)	yes, so hotspot detection works
PocketBase memory	40 MB

13. Phones paired; bug: the app froze on the iPhone with the Pi off

Sreya paired the Android 9 phone ("9lite") and the iPhone; both appear in the device list. With the Pi powered off, the app on the iPhone froze: no button did anything.

Cause	When the hub is switched off, a phone's requests are not refused; they wait (nothing answers at that address). (1) The service worker waited for page loads with **no time limit**. (2) Each button waited up to 8 s for the hub, some twice in a row, with no feedback. (3) The status bar checked only every 30 s. The PC test had missed it: a stopped server on the same machine refuses at once
Fix	Page loads get 3 s, then the cached app opens. Requests get 6 s; once the hub is known to be down, requests fail at once until it answers again. The bar checks every 5 s while the hub is down. Signing out with the hub off returns to the name list. "Check" shows "Checking…"
Test	A test server that accepts connections and never answers (like a switched-off hub seen from a phone): bar red after 4.0 s; sign-out 51 ms; reload shows the app at once with the red bar; real hub started again (page not touched): back to "Online (router)" with the name list after 4.3 s
Deployed	Commit `6ede814`; deploy ok (backup integrity ok, no errors)

Note: a phone gets the fixed version the next time it opens the app **while the Pi is on**. Open it, wait a few seconds, then close and reopen once.

14. P0 step 6: setup wizard (FR-1.01-1.06, 1.13, 1.15)

Sreya chose: **a setup code on the hub** for the first device (DL-44).

Hub

File	Purpose
`hub/pb_migrations/1791200007_p0_setup.js`	`devices.paired_via` gains "setup" (reversible)
`hub/pb_hooks/lib/setup.js`, `setup.pb.js`	Setup code (made at startup, by a once-a-minute job and on demand; kept in `pb_data/setup-code`); monitor message `pb_data/console.issue`; `GET /api/chedam/setup/status`, `POST /api/chedam/setup/start`, `POST /api/chedam/setup/step`
`hub/pb_hooks/lib/access.js`	`business.setup_state` changes only through the step endpoint
`hub/system/chedam-console.path` + `.service`	When the hub's message changes, the monitor's login screen re-reads it (`agetty --reload`); `/etc/issue.d/chedam.issue` links to the message
`hub/scripts/setup-hub.sh`, `deploy.sh`	Install the monitor units and link; print the setup code on a new hub
`hub/tests/step6-setup.test.mjs`	38 tests on a hub **without** sample data

The 38 tests cover: - **The code:** made on a new hub, kept across restarts, shown in the monitor message. - **Refused at start:** a wrong code, a short password, a guessable PIN and a bad email. - **Start:** returns the device key and recovery

code once. Afterwards the code file is gone, the monitor message changes and the code can't be reused. - **The new owner:** PIN, password and recovery code all work. The first device is approved and the owner is signed in on it. - **Store records:** one business with Canadian defaults, plus the main location. - **Steps:** - Language, clock and owner are already done after start. - The profile can be edited, but `setup_state` can't be changed directly. Unknown steps are refused. - A skipped step becomes one task (skipping twice doesn't make two), and doing it later closes the task. When all steps are done, setup is complete. - **Who may run it:** a manager can't mark steps, and start is refused once the store is set up. A PIN the owner sets for someone is temporary. - **Audit log:** setup writes are logged as `system:setup`, with no secrets.

App

Screen	What
Set up your store (3 parts)	Welcome + language + setup code; clock check against the hub (warns at 2+ minutes); owner name, email, password twice, PIN twice, device name
Recovery code	Shown once, Print button, "I have printed it or written it down" required
Wizard	Progress chips (✓ done, ↷ skipped), resumes at the first unfinished step, every step has Save or Skip
Business profile	Store and legal name, type, address, phone, email, GST/HST, PST, BN, CAD, time zone, fiscal year start, prices with or without tax; the main location gets the same address
Logo and receipt	Logo upload (up to 2 MB), colours from the logo (editable, readability warning), receipt header and footer, live receipt preview
Your team	Add people with role, first PIN (temporary), phone, email
Storage areas	"Add the usual four", add with temperature range, mark not used
Features	Ticked from the shop-type preset, change any; core always on, later ones not available
References	Bank name, processor and merchant ID, CRA payroll, CBSA importer, accountant; warning: no account numbers or passwords
Backups	Explained; skip only (adds the task) until step 7
Home	"Finish setting up your store (N steps left)", task list with "Do it", Store setup tile, "Print a new recovery code" (owner)

Browser check (brand-new local hub, no sample data)

Check	Result
Hub start	`setup-code` and the monitor message made at startup
First screen	"Set up your store · 1 of 3"; no sideways scrolling at phone width
Wrong code	Refused at the end with a clear message, back to part 1
Clock check	"OK (under 2 minutes)"
Right code typed in lower case without the dash	Owner created; this browser paired as "Owner's computer"; recovery code shown once; Continue locked until confirmed
Business profile	Saved; wizard moved to Logo and receipt (4 of 10)
Logo	Test logo in orange `#e8701a` and navy `#1b2f5e`: both found exactly; button colour darkened to `#b45714` for readable white text; receipt preview with logo, name, address, GST number, footer
Team	Cashier added with a temporary first PIN
Features	Convenience preset ticked Promotions, Labels, Vendors and Purchasing, Regulated Items, Operations
Skips	References and Backups skipped → 8 of 10 done, two tasks on Home with "Do it"
Saved data	Logo file, colours, receipt text, GST number in the database; logo loads without sign-in (200 image/png)

Failed cases and fixes

Symptom	Cause	Fix

Setup routes answered 404	`onServe` does not exist in PocketBase JS hooks, so the whole file failed to load	`onBootstrap` (after start-up) plus a once-a-minute job and on-demand from the status endpoint
A `sed` edit broke a line in `setup.js`	The inserted comment swallowed the rest of the line	Fixed by hand
Heredoc test script failed in bash	Quotes inside the heredoc	Wrote the test file directly
Svelte warnings "captures the initial value of business"	The forms are meant to start from the saved values once	`untrack()` makes it explicit
Storage: Shelf showed "0 to 0 °C" and the four areas were in A-Z order	PocketBase number fields have no "empty" (they save 0); all four got the same sort number	0 to 0 shows as "room temperature"; each area gets its own sort number

On the Pi (after deploy)

Check	Result
Deploy	Backup integrity ok; `1791200007_p0_setup.js` applied; monitor units installed; no errors
Monitor message	`console.issue` = "Chedam hub: https://\4" (this dev hub has an owner, so no setup code); `/etc/issue.d/chedam.issue` linked; `chedam-console.path` active
`/api/chedam/setup/status`	`needs_setup: false`
PocketBase memory	41 MB

The setup code on the **monitor** will be seen for real on a brand-new hub, for example the spare-card restore test in step 7/10. The dev hub keeps its sample store.

15. P0 step 7: backups (FR-1.14, NFR-03, NFR-04, NFR-12)

Sreya's plan: use her hard drive as the second drive in the testing phase. Production: OTG adapter plus a spare SD card.

Design (DL-48 to DL-52)

Part	What
Root helper `hub/system/chedam-backup` (Python, installed to `/opt/chedam/bin`)	Actions: `drives`, `key-init`, `key-show`, `run`, `test-restore`; technician-only `restore FILE --key KEYFILE`
Hub ↔ helper	Hub writes `pb_data/backup/request.json`; `chedam-backup.path` runs the helper (low priority: `Nice=10`, idle I/O); answer in `result.json`
One backup	SQLite online snapshot → `PRAGMA integrity_check` → gzip → **age** encryption → `Chedam-Backups/<store>/chedam-YYYYMMDD-HHMMSS-<kind>.db.gz.age` + `.sha256` + `.ca` (HTTPS CA, encrypted) → read back from the drive, decrypt, check again, compare table counts = **verified**
Retention	Newest per day 14, per week 8, per month 12, plus the newest 3; only Chedam's own files
Key	age X25519 key in `/opt/chedam/keys` (root only); owner prints it; owner can show it again
Hub (`lib/backup.js`, `backups.pb.js`, migration … `08_p0_backups`)	Status, drives, choose drive, key (owner), run, restore test (owner), nightly schedule (`backup.schedule`, retry after 1 h, 3 per day), lost-job detection (30 min), one urgent task on failure closed by the next good backup, first backup completes the setup step
App	Wizard Backups step: drive → key (print, confirm) → first backup. Backups screen: last good backup, back up now, restore test, drive and nightly time, backup key, recent backups. Home tile
Scripts	`deploy.sh` and `setup-hub.sh` install `age`, the helper and its units

Sreya's hard drive

Drive	Seagate ST31500341AS (Barracuda 7200.11, about 2009), 1.5 TB, NTFS, label "SCRAPP", in a USB enclosure (XT-U33502) with its own power; 1.07 TB free
Pi power	Clean the whole time (`throttled=0x0`); a 3.5" drive has its own adapter
Problem	4 read errors, each after a 30 s timeout and a USB reset, at sectors 6,229,920 to 6,230,944 (about 3.2 GB into the disk). On NTFS that is near where the Master File Table usually sits, and writing any file updates that table
Decision	**Nothing was written to the drive**, to protect Sreya's files. It was only mounted read-only once (to read free space) and unmounted. Advice: copy anything important off it on Windows and check it there (`chkdsk`); use a healthy USB stick or drive for Chedam backups

Failed cases and fixes

Symptom	Cause	Fix
Restore test got 409 "A backup job is still running"	A job marked failed as "lost" (no answer for 30 min) left the in-memory busy flag set	The flag is cleared with the lost job, and a run counts as busy only while its record is still running (test caught it)
Test "request went to the helper" failed	It checked before the fake helper's 100 ms poll	Wait 400 ms
Listing drives took **2 minutes** with the hard drive	The listing mounted each drive to read free space; the sick drive took minutes to mount	The listing never mounts (now 0.55 s); free space comes from each backup
A stuck mount could hold up the helper indefinitely	No time limit	90 s limit; reports "The drive is not responding"
The test image was not listed	Loop devices have type "loop"	Allowed for test images only (`CHEDAM_BACKUP_ALLOW_LOOP=1`)
Test script lost the image ID	`set -e` stopped before saving it; `losetup` needs `sudo`	Looked up by file
The pre-restore copy was readable by every account on the Pi	`copy2` kept root 644	`chedam-hub`, 0640
A rebuilt hub would get a new HTTPS certificate authority (every phone reinstalls)	Not part of the backup	Each backup saves the CA (encrypted); restore puts it back
step2 test failed once: "unknown shorthand flag: 'O'"	The random test password started with "-" (about 1 in 64) and was read as a command option	Test passwords start with "Pw" (both harnesses)

Tests

PC: `node hub/tests/run-all.mjs` → **281** (step7-backups 30/30, with a fake helper that answers like the real one).

On the Pi, the real helper with a 256 MB disk image (ext4) as the drive:

Check	Result
Drive list	0.55 s; lists the image and Sreya's drive without mounting
Key	Made; secret never printed in the terminal
Backup	**1.1 s**, 44 KB encrypted; verified (read back, decrypted, integrity ok, same counts: 5 users, 100 events, …)
On the drive	Only `Chedam-Backups/demostore/…age`, `.sha256`, `.ca`; file starts with `age-encryption.org/v1`; `sha256sum -c` OK
Restore test	Consistent, 0.2 s
Real restore (`chedam-backup restore … --yes`)	**2-3 s** including the hub restart; health ok; integrity ok; same data; pre-restore copy kept

HTTPS after restore	Same CA fingerprint (12:4B:3C:BF…); the laptop with this morning's `ca.crt` connects without change
Hub-to-helper path	A request written as `chedam-hub` was picked up by the path unit and answered
Power	`throttled=0x0` throughout; PocketBase 39 MB

Restore procedure for a dead hub: `docs/P0-backup-restore.md`.

16. Backup card, sign-in mix-up, certificates, 3 devices signed in

New backup card

Sreya connected a 7.4 GB SD card in a USB reader ("Storage Device", FAT32, UUID A8D7-32C4). Read-only check: the first 64 MB read in 5 s with no errors. Power stayed clean (`throttled=0x0`). The card is supported; nothing has been written to it yet. The first backup is made from the app (Backups → drive → key → Back up now → Test a restore).

Sign-in mix-up (not a logic bug)

What Sreya saw	Cause (from the hub's records)	Fix
iPhone at `https://chedam.local` showed a "registration" page	The iPhone was **not paired**: its pairing code ("13pro") was made but never used, so the app showed "Pair this device"	New pairing code from the owner on the laptop, scanned with the iPhone
Owner email + password refused on the iPhone and the laptop	**Demo Owner never had a password**: the dev sample store gives every person a random, unknown password (0 password changes in the audit log). The file `pb_migrations_dev/1791200101_dev_sample_pins.js` holds **PINs** for the name + PIN screen, not passwords	Sreya set a password in the admin UI (users → Demo Owner)
"Locked"	5 wrong tries locked the owner for 15 minutes (`pin_locked_until` 00:53 UTC), the lockout working as designed (NFR-11)	Cleared `pin_locked_until` in the admin UI (or wait 15 min)
Sample people had wrong tries (Demo Owner 2, Cal Cashier 4)	Guessed PINs	Use the PINs from the dev seed file

Lessons for the docs and for later: - The dev sample owner needs a known way in. A dev hub's owner password is set in the admin UI (`/_/`). The setup wizard does this properly for real stores. - Telling apart "not paired", "no password set" and "locked" is hard for the user. The screens could be clearer (for example "This device is not paired yet" in place of a page that looks like registration). Noted for the step 5/6 polish.

Certificates on the three devices

Device	Before	After
Android 9 (9lite)	Trusted (paired, HTTPS since the morning)	Same
Surface laptop (Windows, Chrome)	**Not trusted**: Chrome showed "Not secure" with `https` crossed out on `https://192.168.50.101/_/`	Sreya installed `ca.crt` into Current User → Trusted Root Certification Authorities and restarted Chrome
iPhone 13 Pro (iOS 26.6.1, Chrome 154)	Profile installed; trust toggle needed checking	Certificate Trust Settings → Chedam Store CA on; paired

Test for any device: `https://192.168.50.101/hub.json` opens with no warning.

Result: 3 devices signed in (P0 gate item "3 devices log in")

Device	Hub record	Browser (from the hub)	Signed in
iPhone 13 Pro	"13pro", approved, app 0.5.0	iOS 26.6.1, Chrome 154 (CriOS)	Demo Owner

Android 9 phone (Honor 9 Lite, LLD-AL10)	"9lite", approved, app 0.5.0	Android 9, **Chrome 98**	Demo Owner
Surface laptop	Owner on any device (not paired, FR-1.09)	Chrome on Windows	Demo Owner

Two earlier iPhone pairing attempts ("13pro", "13pronew") were removed in the device manager (status revoked), as intended.

Issue: Android 9 phone shows misaligned screens

Symptom	Screens work but are not properly aligned on the Android 9 phone
Cause	The phone's Chrome is version **98** (February 2022). The app uses Tailwind CSS 4, which needs Chrome 111+ (cascade layers, `color-mix()`, `oklch` colours). Chrome 98 ignores those rules, so part of the styling is lost. My assumption that Android 9 would have Chrome 138 was wrong (risk note in the mini-spec)
Options	(a) Update Chrome on that phone from the Play Store (Android 9 still gets Chrome up to about 138), which fixes it at once if updates work. (b) Make the app's CSS work on older Chrome too (Tailwind 3.4 or a compatibility build), the safer choice for pilot stores with old phones
Status	Decided (DL-53): no extra work; Sreya will test on another, up-to-date phone

17. P0 step 8: hub health page (FR-12.09)

What it shows

Item	Source (read as the hub's sandboxed user)	Status rule
Hub running for	`/proc/uptime`, model from `/proc/device-tree/model`	info
Temperature	`/sys/class/thermal/thermal_zone0/temp`	warn ≥ 70 °C, bad ≥ 80 °C (the Pi slows down)
Power supply	`rpi_volt` sensor `in0_lcrit_alarm` (under-voltage now)	bad if too low
Hub app memory	`/proc/self/status` VmRSS	warn > 150 MB (NFR-08)
Free memory	`/proc/meminfo` MemAvailable	warn < 100 MB, bad < 50 MB
Free storage / store data	`df` on `pb_data`; size of `data.db` + WAL	warn < 15 %, bad < 5 %
Clock	`chronyc -c tracking` (stratum, offset, leap); `/dev/rtc0`	warn if not synchronised or > 2 s off; notes "no battery clock (RTC)" (DL-23)
Internet	connectivity status (FR-12.01)	info
Last backup	backups status	bad if no drive or > 50 h; warn > 26 h (NFR-04)
Last restore test	`backup.last_restore_test`	bad if failed; warn > 35 days (NFR-03)
Devices online	device manager view	warn if devices wait for approval
Waiting to sync	offline queue arrives with selling (P1)	info
Versions	app build (`sw.js`), schema (newest non-dev migration), PocketBase, Caddy, OS, kernel	info
Pending updates	`updates` table (channel in step 9)	warn if one failed

- The top line shows the worst status ("All good" / "Needs a look" / "Needs attention now"). Answers are cached for 20 s, and "Check now" forces a fresh check.
- Endpoint: `GET /api/chedam/health` (health.view: owner, manager). App: Hub health screen (refreshes every 30 s, links to Backups and Devices), a home tile with ✓ / ! / ✗, and a red "The hub needs attention" card on the home screen.

Checks

Check	Result

Which readings the hub user can access on the Pi	All, except `vcgencmd`/throttling flags (needs root); the `rpi_volt` sensor gives under-voltage instead
Same commands **inside the hub's systemd sandbox** (`systemd-run` with the service's restrictions)	chrony: synced, stratum 2; df: 117.7 GB free of 125.4 GB; 40.8 °C; power alarm 0; Caddy 2.6.2; PocketBase 0.40.4
Tests `step8-health.test.mjs`	21/21: 401 signed out, 403 cashier, every item present with a status, no backup drive → bad, 30 h old backup → warn, fresh → ok, restore test "never", schema shown, cached answers
All tests	**302** passing
Deploy	Backup integrity ok; no errors; endpoint answers 401 without sign-in

Failed cases and fixes

Symptom	Cause	Fix
"Schema" showed `1791200101_dev_sample_pins`	The newest migration was the dev sample-data one	Ignore files containing `_dev_`
The fix then showed "Schema ?"	Backslash escaping through Python → JS → SQL `ESCAPE` broke the query (the same trap as on day 2)	`instr(file, '_dev_') = 0`, no escaping needed

18. P0 step 9: updates (FR-12.02, FR-12.04, NFR-20)

Sreya created the public repo `repatikosamid-prog/chedam-releases` (DL-54). The full guide is in `docs/P0-updates.md`; the decisions are DL-55 to DL-58.

What was built

Part	What
Signing key	Ed25519 made with OpenSSL in `~/.chedam/update-signing.key` (outside the repo; Windows permissions: Sreya's account only); public key `hub/system/update-signing.pub`
`tools/release/make-release.sh`	Build the app → stage hub code (refuses dev migrations) → `manifest.json` (version, notes, commit, migrations) → tar.gz → sign → check the signature against the repo's public key → signed `stable/channel.json` in `../chedam-releases`; `--publish` pushes
`hub/system/chedam-update` (root) + `.path`/`.service`	`check` (https channel, signatures, checksum), `scan` (USB `chedam-update/` and `/opt/chedam/updates/inbox`), `install` (signature → DB snapshot + code copy → swap → restart → health or automatic rollback)
Hub: migration … `09_p0_updates`, `lib/updates.js`, `updates.pb.js`	Status, check online/USB, install now (owner) or in the 02:00–05:00 window, results read after the restart, an urgent task on failure, a daily background check when online (not blocking)
App	Updates screen (installed version, check online / USB stick, release notes, "signed ✓", install now / tonight, history), home tile; product version 0.9.0
Scripts	`deploy.sh`/`setup-hub.sh` install the helper, units and public key; `/opt/chedam/keys` stays root-only (it also holds the backup key); a deploy writes `VERSION` = `dev-<commit>`
Local clone	`C:\Users\Venkata\Desktop\Projects\chedam-releases` with a README

Tests

PC: `step9-updates.test.mjs` 21/21 with a fake helper: - **Check:** cashier refused; online check finds 2 updates; the channel address is passed; no duplicate records; the offline check is recorded. - **USB scan:** finds the package. - **Install:** manager refused, owner installs now; a second install is refused while one runs (409); the result is read after the restart. - **Rollback:** reported, with an urgent task. - **Install window:** scheduled → installed when the window opens; the task closes. - **Daily check:** sent without waiting, its answer recorded at the next tick. - **Audit log:** installs and rollbacks are logged.

All suites: **323**.

On the Pi with the real helper and real signed packages (local, not yet published):

Test	Result
Package 0.9.0	118,545 bytes, 44 files, no dev data, keys or databases; signature ok
Scan of the inbox	Found 0.9.0 (signed)
Install 0.9.0 over `dev-cd633e3`	**1.7 s** (3 s end to end); VERSION 0.9.0; app build `0.9.0-15ad57cd6968`; 5 users and 129 events unchanged; integrity ok; dev migrations kept
Broken 0.9.1 (signed; migration throws on purpose; built in the scratch folder only)	First run: rolled back after **93 s** (waited out the 90 s limit). After the fail-fast fix: rolled back after **4 s**; version 0.9.0; data and integrity unchanged; broken migration never applied
Good 0.9.0 again (`--force`) after the fixes	1.6 s, healthy
Tampered package (1 byte added)	**Refused**: "The signature does not match"; nothing changed

Failed cases and fixes

Symptom	Cause	Fix
Permission command for the key failed (`Invalid parameter ":USERNAME:(R,W)"`)	PowerShell variables written inside a Bash command	Ran it in PowerShell; the key is now readable by Sreya's account only
`deploy.sh`/`setup-hub.sh` would have opened `/opt/chedam/keys` to 755	Copy-paste of the updates folder line	Kept at 700 (the backup key lives there)
Hub test: second install was not refused; test crashed	The fake helper finished in 0.2 s, so the first install was already done	The fake install takes 1.5 s; the 409 is checked at once
Daily check would block a hub job for up to 2 min	It waited for the helper's answer	It now only sends the request; the answer is read at the next tick
`make-release.sh`: "Cannot find module '/c/Users/…/package.json'"	Node on Windows cannot read Git Bash paths written inside a script	Read the version with a relative path
Building the broken test package: "Cannot connect to C: resolve failed"	GNU tar takes `C:` as a remote host	Used `/c/…` paths
Broken update took 93 s to roll back	The installer waited the full 90 s although the hub logged "failed to apply migration" after 3 s	It watches the log while waiting and rolls back at the first crash
After that fix the rollback was reported as "did not come up" although it had worked	The rollback check read the log "since this second", which still held the crash lines from the same second	Journal **cursors**: each check reads only lines written after it started

19. First real backup verified; 0.9.0 published

Backup to the new SD card (made by Sreya in the app)

Check	Result
Records	2 backups, both **verified**: a nightly one, which started by itself the moment the drive and key were ready (tonight's slot had not run yet, the catch-up rule), and the first backup from the wizard (48,118 bytes); 7.9 GB free
Setup task "choose a backup USB drive"	done
Sreya's restore test in the app	passed, consistent, 0.3 s
Independent check (card mounted read-only)	Only Chedam's folder added (plus Windows' "System Volume Information"); `sha256sum -c` OK for both; both files start with `age-encryption.org/v1`; `.ca` files present
Independent restore test (helper)	Newest backup rebuilt in 0.3 s, consistent (5 users, 4 devices, 120 events, …)
Power	`throttled=0x0` (the 4 I/O errors on record are the old hard drive's)

Publishing 0.9.0 (with Sreya's OK)

```
bash tools/release/make-release.sh --notes "Chedam 0.9.0 (P0): …" --publish
```

Check	Result
Package	`chedam-app-0.9.0.tar.gz`, 118,582 bytes, sha256 `d6f0b698…`, built from committed code, signature ok
Public repo `chedam-releases`	Commit `3c87ec6`: README, `stable/channel.json(.sig)`, `packages/chedam-app-0.9.0.tar.gz(.sig)`
Pi downloads from GitHub (`chedam-update check`)	Channel and package signatures valid, checksum matches; 0.9.0 found as newer than `dev-f632959`

Issue found and fixed

Symptom	Cause	Fix
Git warned "LF will be replaced by CRLF" for files in the releases repo	Git on Windows may convert line endings in files it takes for text. A 64-byte signature has no zero byte about 78% of the time, so Git can treat it as text, and changing a single byte breaks the signature. This time the files arrived intact (verified by the Pi)	`.gitattributes` with `* -text` in `chedam-releases` (commit `ba25f3c`); `make-release.sh` refuses to run without it

20. P0 gate: memory under load (NFR-07, NFR-08, risk R6)

Method

Part	What
Test hub	`tools/loadtest/pi-loadtest.sh`: a **throwaway** hub on the Pi (same PocketBase binary, hooks and migrations including the sample store; its own data folder; 127.0.0.1:8099 only), started with the **same limits as the service** (GOMEMLIMIT 100 MiB, MemoryHigh 150 MB, MemoryMax 220 MB), plus a sampler (RSS/peak every 2 s, load, free memory). The store's real hub was not touched. Deleted afterwards
Load	`tools/loadtest/hub-load.mjs` on the laptop through an SSH tunnel: **10 devices** (twice the pilot's 5, NFR-07), each paired and signed in by PIN, every 1-2 s: connectivity, device status, my permissions, settings, modules, storage areas; writes (create + update, each also writes an event) by the roles allowed to; a PIN sign-in every 10 rounds (bcrypt); health page; plus every 30 s a **burst** of simultaneous audit-log reads (100 events per page)
Duration	5 minutes per run

Results

Run	Requests	Errors	Latency p50 / p95 / p99	Hub memory peak (VmHWM)	Pi free memory (lowest)	NFR-08
1 (script mistakes, see below)	12,729	1,106	34 / 136 / 530 ms	86 MB	174 MB	(not valid)
2: burst of **50**	10,805	0	37 / 766 / 5,034 ms	**197 MB**	92 MB	**FAIL**
3: burst of **10** (all devices at once: realistic)	11,723	0	35 / 94 / 552 ms	**108 MB**	174 MB	**PASS**
4: burst of 50, **with rate limits**	11,961	360 × 429 (bursts only)	36 / 470 / 920 ms	**120 MB**	162 MB	**PASS**

The real hub afterwards: 43 MB, healthy through Caddy.

Fix: per-device rate limits (DL-59)

- Migration `1791200010_p0_rate_limits.js`:
- limits: audit-log lists 10 per 3 s, all lists 40 per 3 s, PIN sign-in 15 per 10 s, all API calls 150 per 3 s;
- the device's address comes from `X-Forwarded-For` (Caddy sets it), and 127.0.0.1 is exempt.
- Without the trusted header, every device would look like 127.0.0.1 behind Caddy and share one allowance.

- Test `step10-ratelimit.test.mjs` (5/5):
- a flooding device gets about 10 answers, the rest 429;
- another device is not affected, and the flooder is served again after 3 s;
- a busy till (36 requests in 3 s) is never limited;
- 127.0.0.1 is exempt.
- All suites: **328**.

Failed cases and fixes

Symptom	Cause	Fix
Run 1: 1,106 errors	Test script: cashier and accountant tills tried to add storage areas (correctly refused, 403); the bursts sorted the audit log by `created_at`, but its field is `at` (400)	Writes only by roles allowed to; sort by `at`. The run-1 memory figure was not valid: the bursts had failed instantly
Run 2: 197 MB, p99 5 s	50 simultaneous audit-log reads: each SQLite connection keeps its own page cache, and PocketBase has no option to cap connections	Per-device rate limits (DL-59)
"Through Caddy" check on the Pi failed (curl exit 77)	My Pi account cannot read Caddy's certificate file	Checked from the laptop: 200

Next

1. Sreya: install 0.9.0 from the Updates screen (steps in the chat).
2. Gate tests left: **power pull ×10** mid-write (Sreya pulls the plug, I run the writes and the checks) and **camera over HTTPS** on the iPhone and an up-to-date Android phone.

2026-10-03 (Saturday, October 3, 2026)

New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed

Source: `docs/P0-pi-setup-log-2026-10-03.pdf` (detailed log of this day) and `docs/P0-restart-new-network.md` (the plan). This entry carries its content forward.

Summary

- The TELUS modem gives no admin access, so DHCP reservations are impossible. The dev hub moved behind our **own TP-Link router**, which is LAN-cabled to the bedroom wall port.
- Network: subnet `192.168.50.0/24`, gateway `192.168.50.1`, 2.4 GHz SSID `TP-LINK_2CBE`.
- Pi `chedam` at **192.168.50.101** (MAC 88:A2:9E:5C:17:65, reservation enabled); dev PC DESKTOP-AHG4TD7 at 192.168.50.100.
- The SD card was wiped and re-flashed. Key-only SSH works, and the OS is updated.
- HDMI console made visible. `setup-hub.sh` gained the `--keep-display` option.
- Commits: `ea0a7e6`, `678eee4`, `bacf585`.

Why SSH keys instead of a password

With a key pair, the private key never leaves the PC and the Pi holds only the public key. No secret crosses the network, keys cannot be guessed, and scripts can log in, so password login is switched off.

File	Where	Rule
`id_ed25519`	PC `C:\Users\Venkata\.ssh\`	Private. Never share, paste or commit; back it up privately
`id_ed25519.pub`	PC, same folder	Public, one line ending `chedam-dev`. Safe to copy
`authorized_keys`	Pi `/home/chedam/.ssh/`	Allowed public keys; folder 700, file 600
`known_hosts`	PC `.ssh\`	Remembered Pi identities; `ssh-keygen -R <ip>` after a re-flash
`config`	PC `.ssh\`	Alias `ssh chedam`

Key procedures recorded:

- create the key pair
- put the public key on the Pi with Imager (REMOVE old keys, then BROWSE to the `.pub` file)
- add or replace keys with `type new.pub | ssh chedam "cat >> ~/.ssh/authorized_keys"`
- what to do when locked out (monitor + keyboard, or re-flash)

Steps

1. Backed up `.ssh` with `Copy-Item $env:USERPROFILE\.ssh $env:USERPROFILE\.ssh_backup -Recurse`.
2. Cleared old host fingerprints by deleting `known_hosts` and `known_hosts.old`. There was a backup.
3. Router: changed the admin password, set the LAN to `192.168.50.1/24` (different from TELUS `192.168.1.x`), kept it in router mode, and turned on 2.4 GHz.
4. Re-flashed the card. Imager: Pi Zero 2 W, Lite 64-bit, `chedam`/`chedam`, new Wi-Fi, CA, SSH public-key only.
5. Found the Pi in the router's DHCP client list and created the reservation.
6. Updated the SSH config to `HostName 192.168.50.101` and removed `HostKeyAlias`.
7. Ran `sudo apt update && sudo apt full-upgrade -y` and checked that `sshd -T` shows `passwordauthentication no`.
8. Readiness check: model, OS, IP, active Wi-Fi, `multi-user.target`; reboot test passed.
9. Forced HDMI output (see below).

Failed cases and fixes

Attempt / symptom	Cause	Fix
`ssh-keygen -R <old-ip>` errored	PowerShell treats `<` as an operator	Use real addresses; deleting `known_hosts` avoided needing the old IP
Key pair deleted during "start fresh"	Not needed: a key is not tied to a network	Restored from `.ssh_backup`
`ssh 192.168.50.101` denied as `venkata`	No user given, so the Windows name was used	`ssh chedam@192.168.50.101`
`ssh chedam@chedam` timed out	Windows cannot resolve a bare name	IP, `chedam.local`, or the alias
`-i $env:venkata\.ssh\...` not found	`$env:USERPROFILE` is a real variable, not a placeholder	Keep `$env:USERPROFILE`
`Permission denied (publickey)`	Imager wrote an old remembered key (… `COL3o srey…`), not the PC's (… `LuhpI0dY chedam-dev`)	Re-flash; REMOVE old keys, BROWSE to `id_ed25519.pub`
A command pasted into Imager's key box	`Get-Content … | Set-Clipboard` text pasted as the key	REMOVE, then BROWSE to the file
"REMOTE HOST IDENTIFICATION HAS CHANGED"	Expected after a re-flash (new host key)	`ssh-keygen -R 192.168.50.101`
Reservation saved as *Disabled* (TP-Link)	Router default	Tick it, then Enable Selected; leave ARP binding off
HDMI blank	`dmesg`: `vc4-drm: Cannot find any crtc or sizes`	Append `video=HDMI-A-1:1280x720@60D` to the single line in `/boot/firmware/cmdline.txt`, then reboot
Lines after `sudo cp` lost	The sudo password prompt ate the pasted lines	Paste one command at a time when sudo may prompt

HDMI fix commands:

```
sudo cp /boot/firmware/cmdline.txt /boot/firmware/cmdline.txt.bak
sudo sed -i '1 s/$/ video=HDMI-A-1:1280x720@60D/' /boot/firmware/cmdline.txt
sudo reboot
```

Lessons

- Before flashing, check that the key shown in Imager ends the same as `type C:\Users\Venkata\.ssh\id_ed25519.pub`. Imager remembers old keys.
- Always back up `.ssh` before changes.
- With password login off, losing `id_ed25519` means re-flashing or using a keyboard and screen.
- No passwords or key contents are stored in the repo.

2026-10-02 (Friday, October 2, 2026)

P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

Summary

- Read Master Specification v1.0 and pulled out the P0 (Foundation) scope, done-criteria and phase gate.
- Wrote `docs/P0-readiness-checklist.md`: hardware, software with versions and links, Pi flashing steps, accounts.
- Installed and verified the developer tools on the Windows PC (all checksums matched the official ones).
- Flashed the Pi. The first attempts failed on SSH keys (including the private key being pasted into Imager); the key was replaced and SSH worked.
- Wrote the hub setup (`hub/scripts/setup-hub.sh` + `hub/system/*`) and ran it on the Pi at `192.168.1.241` (TELUS network). HTTPS worked by name and IP.
- Wrote `docs/P0-mini-spec.md` (P0 build order, decisions DL-21 to DL-25).
- Decisions from Sreya: RTC skipped for now; cloud backup tested later when client IDs exist; test phones are Android 9 and iPhone 13 Pro (iOS 26.6.1); update channel option A (source stays private, signed packages in a separate public repo).

1. Spec review

Source: `SDLC Docs Toosl softweare installers and ReadMe/MASTER Specifications/Chedam — Master Specification v1.0.pdf` (41 pages).

The PDF text was extracted with `pdftotext -layout`. The release-plan and architecture pages are images, so they were rendered with PyMuPDF and read as pictures.

P0 requirements: FR-1.01 to 1.10, 1.13, 1.14, 1.15; FR-12.01 to 12.04, 12.08, 12.09.

Done when: 3 devices log in, the hub survives a power pull, an update installs and rolls back. The phase gate adds: memory under 150 MB (NFR-08), a load test (R6), an iOS/Android HTTPS camera test (R5), and a verified backup and restore.

2. Developer PC tools

Already present: Git 2.52.0, VS Code 1.135, Python 3.13, Windows OpenSSH.

Tool	Version	Verified against	Result
Node.js	24.21.0 (LTS Krypton)	nodejs.org SHASUMS256 + OpenJS signature	Match, installed
PocketBase (Windows)	0.40.4	PocketBase `checksums.txt`	Match
DuckDB CLI	1.5.6	GitHub release digest	Match
Raspberry Pi Imager	2.0.11.1	GitHub digest + Raspberry Pi Ltd signature	Match, installed
DB Browser for SQLite	3.13.1	winget manifest + SignPath signature	Match, installed
mkcert	1.4.4	Scoop manifest (file is unsigned)	Match
GitHub CLI	2.102.0	n/a	Installed, logged in as `repatikosamid-prog`

Commands used for verification (Git Bash):

```
sha256sum *.msi *.exe *.zip
curl -s https://nodejs.org/dist/v24.21.0/SHASUMS256.txt | grep x64.msi
```

```
powershell Get-AuthenticodeSignature <file>
```

Failed case: `mkcert` not recognized

- Symptom: `mkcert -install` → "The term 'mkcert' is not recognized".
- Cause: the file was still named `mkcert-v1.4.4-windows-amd64.exe`, and `C:\Tools` was not on PATH. PowerShell also never runs a program from the current folder without `.\`.
- Fix: copied `mkcert.exe`, `pocketbase.exe` and `duckdb.exe` to the top of `C:\Tools` and added `C:\Tools` to the user PATH. A new PowerShell window was needed for the change to show.

Then Sreya ran `mkcert -install` (local dev CA added to the Windows trust store) and `gh auth login` (fine-grained token).
The installers folder was added to `.gitignore`. PocketBase had created test data there.

3. Flashing the Pi (first time)

Imager settings: Pi Zero 2 W, Raspberry Pi OS Lite 64-bit, hostname `chedam`, user `chedam`, Wi-Fi 2.4 GHz, country CA, time zone America/Vancouver, SSH public-key only.

Failed case 1: hostname and username

- The hostname was typed as `Chedam` and the username as the GitHub name.
- Fix: lowercase `chedam` for both, so every script and guide matches.

Failed case 2: no SSH key existed

- SSH was set to password authentication, and the PC had no key yet.
- Fix: `ssh-keygen -t ed25519 -C "chedam-dev"`.

Failed case 3: `Permission denied (publickey)`

- Debug output showed the PC offering its key and the Pi refusing it for both usernames:

```
ssh -v -o BatchMode=yes chedam@chedam.local true
```

- Cause: the key written to the card was not the PC's key.
- The "host key differs" warning came from an old `192.168.1.241` entry in `known_hosts`. It was removed with `ssh-keygen -R`.

Failed case 4 (security): private key pasted into Imager

- Imager listed 7 "keys". They were the 7 lines of the **private** key (`BEGIN OPENSSH PRIVATE KEY`).
- Risk: the private key was copied onto the SD card and partly shown in a screenshot.
- Fix: removed all 7 entries, deleted the key pair, generated a new one, and pasted **only** `id_ed25519.pub` (one line, `ssh-ed25519 … chedam-dev`). Re-flashing overwrote the card.

Failed case 5: `.local` name not resolving from Windows tools

- Git Bash ssh and Windows ssh both failed on `chedam.local`, while `Resolve-DnsName` worked.
- Fix: an SSH alias `chedam` pointing at the IP, in `~/.ssh/config`. The Pi first booted slowly (~5 min), so early checks also failed.

Login worked: `chedam@chedam`, Debian 13 trixie, aarch64, 415 MB RAM, 117 GB card.

Admin access

`sudo` asked for a password, so Sreya ran this once:

```
ssh -t chedam "echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmo
```

4. Hub setup (first time, Pi at 192.168.1.241)

OS update (ran in the background, ~15 min):

```
sudo apt-get update && sudo apt-get -y full-upgrade && sudo apt-get -y install chrony sqlite3 ufw
```

PocketBase 0.40.4 linux_arm64 was downloaded on the Pi and checked with `sha256sum -c` against `checksums.txt`.
Files written in the repo:

File	Purpose
`hub/scripts/setup-hub.sh`	Idempotent setup: packages, service user `chedam-hub`, `/opt/chedam`, PocketBase service, Caddy, watchdog, journald, Wi-Fi power save off, boot config, firewall
`hub/system/chedam-hub.service`	PocketBase on 127.0.0.1:8090, auto-restart, `GOMEMLIMIT=100MiB`, `MemoryMax=220M`, hardening
`hub/system/Caddyfile`	HTTPS for `chedam.local` and the hub IP, using a per-store CA; plain HTTP only for `/ca.crt`
`hub/system/watchdog.conf`	Hardware watchdog 14 s
`hub/system/journald.conf`	Logs capped at 64 MB (SD card wear)
`hub/system/wifi-powersave-off.conf`	Lower Wi-Fi latency
`hub/system/index.html`	Placeholder page "Chedam hub is running"

Run on the Pi:

```
scp -r scripts system chedam:hub/
ssh chedam 'sudo bash ~/hub/scripts/setup-hub.sh'
ssh chedam 'sudo systemctl reboot'
```

Failed case 6: GitHub repo creation

- `gh repo create … --public` → "Resource not accessible by personal access token".
- Cause: the fine-grained token cannot create repositories. Sreya has to create the public repo on github.com.

Failed case 7: HTTPS tests from the PC returned nothing

- Windows `curl` (Schannel) → "the revocation status is unknown". A private CA has no revocation server.
- Fix for command-line tests: `--ssl-no-revoke`. Browsers are not affected.

Failed case 8: Caddy error at start

- Caddy tried to install its root certificate into the Pi's own trust store and failed ("failed to execute sudo").
- Fix: the `skip_install_trust` global option in the Caddyfile.

Results after reboot

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon active
HTTPS	`https://chedam.local/api/health` and `https://192.168.1.241/api/health` → 200
Certificate	Valid against "Chedam Store CA - 2026 ECC Root" (valid to 2036); IP cert has IP SAN
Memory	PocketBase 27 MB + Caddy 41 MB; total RAM rose from 415 to 462 MB after headless config
Watchdog	14 s
Temperature	52.6 °C, no throttling

5. Decisions recorded (docs/P0-mini-spec.md)

ID	Decision
DL-21	HTTPS via Caddy with an internal CA created on each hub (one per store); mkcert is dev-PC only
DL-22	Source repo private; signed update packages in a separate public repo
DL-23	RTC deferred for dev; required for the pilot
DL-24	Headless boot config (later changed to a switch, see 2026-10-05)
DL-25	PocketBase extended with JavaScript hooks; no Go build

Findings:

- Android 9 cannot resolve `.local`, so the hub needs a fixed IP.
- Android 9's Chrome no longer gets updates.
- Caddy leaf certificates last 12 hours, so the clock must be correct (RTC for the pilot).